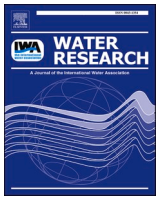
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A novel approach for estimating and predicting uncertainty in water quality index model using machine learning approaches

Md Galal Uddin [a,b,c,*], Stephen Nash [a,b,c], Azizur Rahman [d,e], Agnieszka I. Olbert [a,b,c]

[a] Civil Engineering, School of Engineering, College of Science and Engineering, University of Galway, Ireland
[b] Ryan Institute, University of Galway, Ireland
[c] MaREI Research Centre, University of Galway, Ireland
[d] School of Computing, Mathematics and Engineering, Charles Sturt University, Wagga Wagga, Australia
[e] The Gulbali Institute of Agriculture, Water and Environment, Charles Sturt University, Wagga Wagga, Australia

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ABSTRACT

With the significant increase in WQI applications worldwide and lack of specific application guidelines, accuracy and reliability of WQI models is a major issue. It has been reported that WQI models produce significant uncertainties during the various stages of their application including: (i) water quality indicator selection, (ii) sub-index (SI) calculation, (iii) water quality indicator weighting and (iv) aggregation of sub-indices to calculate the overall index. This research provides a robust statistically sound methodology for assessment of WQI model uncertainties. Eight WQI models are considered. The Monte Carlo simulation (MCS) technique was applied to estimate model uncertainty, while the Gaussian Process Regression (GPR) algorithm was utilised to predict uncertainties in the WQI models at each sampling site. The sub-index functions were found to contribute to considerable uncertainty and hence affect the model reliability – they contributed 12.86% and 10.27% of uncertainty for summer and winter applications, respectively. Therefore, the selection of sub-index function needs to be made with care. A low uncertainty of less than 1% was produced by the water quality indicator selection and weighting processes. Significant statistical differences were found between various aggregation functions. The weighted quadratic mean (WQM) function was found to provide a plausible assessment of water quality of coastal waters at reduced uncertainty levels. The findings of this study also suggest that the unweighted root means squared (RMS) aggregation function could be potentially also used for assessment of coastal water quality. Findings from this research could inform a range of stakeholders including decision-makers, researchers, and agencies responsible for water quality monitoring, assessment and management.

1. Introduction

Uncertainty quantification (UQ) is an essential step in any experimental or computational assessment, as it determines how trustworthy the measured or computed data are; this also applies to water quality assessment. The water quality index (WQI) model is a widely used tool for evaluating water quality (WQ). This simple yet powerful tool facilitates the transformation of large quantities of often-complex inter-correlated WQ data into a single numerical value (Uddin et al., 2017; 2022a; 2023). Its popularity has grown steadily in recent years as a result of its simple architecture, ease of application and straightforward result interpretation. The WQI model is comprises of four components: (1) WQ indicators selection, (2) sub-index (SI) function for converting different units to a uniform scale, (3) parameter weight value generation based on their relative importance, and (4) aggregation function for combining all three components and converting them into a single value (Uddin et al., 2022a). To date, over 20 WQI models have been developed by states or organizations worldwide (Uddin et al., 2021). One common and major drawback of these models is the model uncertainty problem.

There have been many studies that compare various WQI models and assess their predicative abilities. These studies mainly discuss the model structure and the effect of aggregation function on WQI scores. A small number go further and try to quantify the uncertainty introduced by the various steps of a WQI model. It is argued that there are many sources of model uncertainties and they occur at various stages of WQI modelling process (Abbasi and Abbasi, 2012; Uddin et al., 2021). Juwana et al.

* Corresponding author at: Civil Engineering, School of Engineering, College of Science and Engineering, University of Galway, Ireland.
E-mail address: u.mdgalal1@nuigalway.ie (M.G. Uddin).

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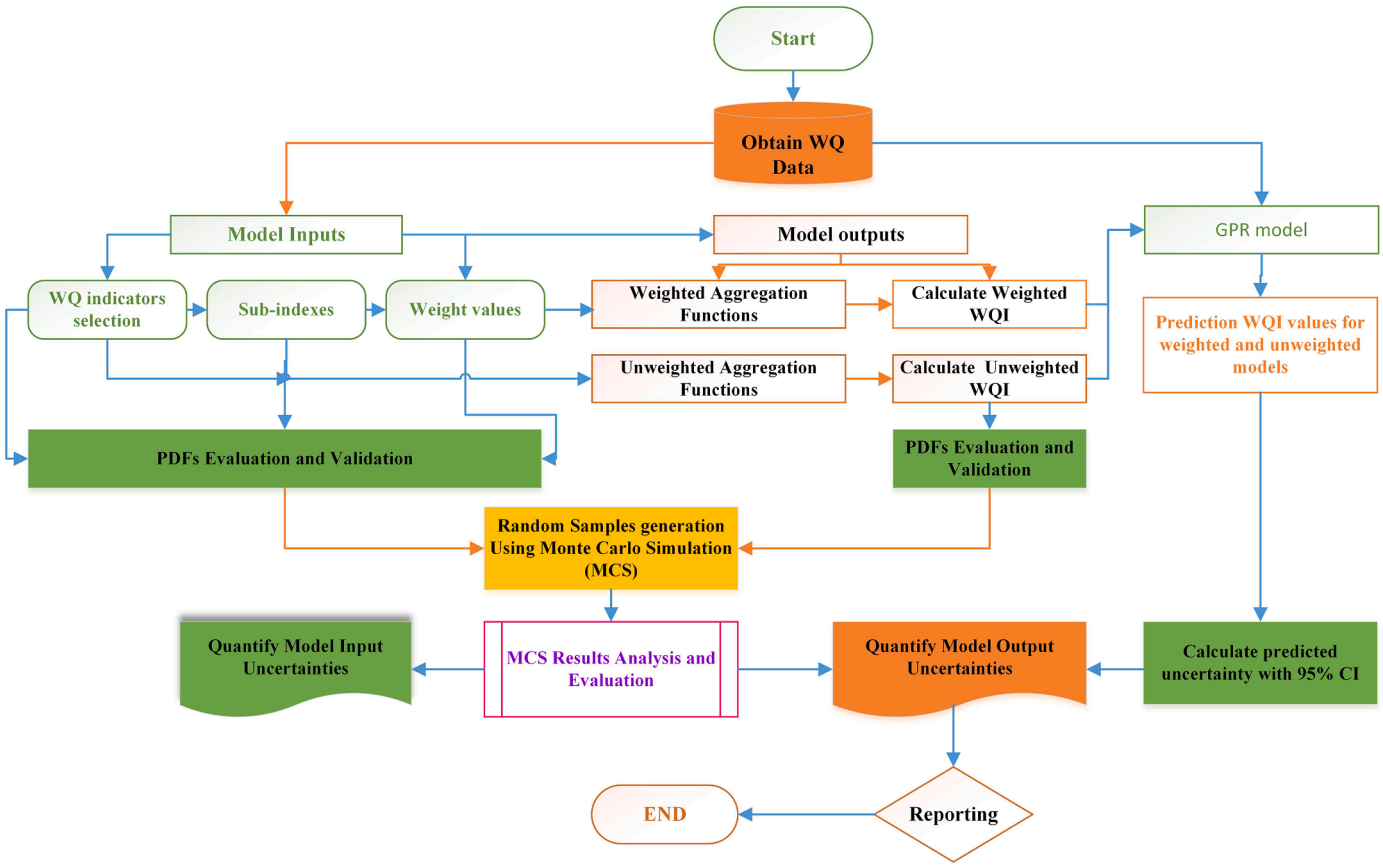


Fig. 1. A schematic methodological framework of Monte Carlo Simulation for uncertainty analysis in WQI model.

(2016) and Sutadian et al. (2017) found that the sub-index functions and weight generation methods may contribute to the generation of model uncertainty; however, Uddin et al. (2022b) implied that the weighting procedure does not have significant effects on model uncertainty. Although recent research concludes that the aggregation function contributes to uncertainty generation (Juwana et al., 2016; Parween et al., 2022; Sutadian et al., 2018), no quantification of uncertainties at this step has yet been performed.

As expected, the structure of aggregation function in the WQI model has also an effect on the uncertainty (Abbasi and Abbasi, 2012; Smith, 1990). Smith (1990) and Lumb et al. (2011) conducted a comparative study of different WQI models, and found a significant variation in a use of unique WQ indicators among different aggregation functions. Uddin et al. (2022a) compared the performance of eight aggregation functions and found a significant statistical difference in WQI scores produced by these functions.

Model uncertainty, and directly linked to it model sensitivity, are the most fundamental features and characteristics of a model. The uncertainty refers to the result of measurement that reflects a lack of exact knowledge of the value of the measurand (Rahman and Harding, 2017; BIPM et al., 2020). In order to understand the model's accuracy and reliability, it is important to estimate the uncertainty (Sankaran and Sarkar, 2009). Four types of uncertainties are typically evaluated in the models: structural, functional, parameter and data uncertainty (Ng and Perera, 2001; Sankaran and Sarkar, 2009; Wu et al., 2006). Uncertainties in WQI models are often related to various components of a model and its processes. Nonetheless, it is unclear how much of the total uncertainty is attributed to each component of the WQI model. Moreover, to the best of our knowledge, there is no specific framework for the estimation of the overall propagation of uncertainties in a WQI model.

In this context, the purpose of this study is to understand the generation of uncertainties along the cascade of WQI model processes, and accurately quantify how much uncertainty is conveyed by each component of the model. For the first time in this study, the individual and combined effects of inputs from steps (1)-(3) to the aggregation function are investigated.

A variety of methods can be reliably utilized for the quantification of model uncertainty (Rahman et al., 2013). Two common techniques widely used are: (i) local and (ii) global approaches. The local technique is adopted to determine the sensitivity of the model's input parameters (Zhao et al., 2011). In contrast, the global method is suitable to estimate a range of possible parameter values in the parameter distribution (Jiang et al., 2018; Mishra, 2009). In this research, the Monte Carlo simulation (MCS) approach is employed to quantify the uncertainties in the WQI model by following the guide to the expression of uncertainty in measurement. GUM is the fundamental reference guide that provides an in-depth description of how to evaluate measurement uncertainty using the MCS method (Joint Committee For Guides In Metrology, 2008; Rodríguez et al., 1989). The MCS method is both systematic and comprehensive in assessing model uncertainty, and as such it is widely used (Jafari et al., 2016; Mishra, 2009).

Advanced artificial intelligence and machine learning (ML) techniques are also widely used to predict model uncertainty (Abdar et al., 2021; Tavazza et al., 2021). These techniques predict the confidence interval (CI) of measured values to determine how much data variation occurs in a dataset. Consequently, the model errors can be quantified and optimized using these techniques (Rahman, 2019). In this study, the Gaussian Process Regression (GPR) ML algorithm was applied to predict WQI values with a 95% confidence interval. This was followed by a comparative analysis between the MCS and GPR results of model uncertainty to ultimately determine the model efficacy.

This paper is divided into four main sections. In the first section, a brief overview of this study is provided. Section two discusses a new set of tools and techniques that are applied to estimate and predict model

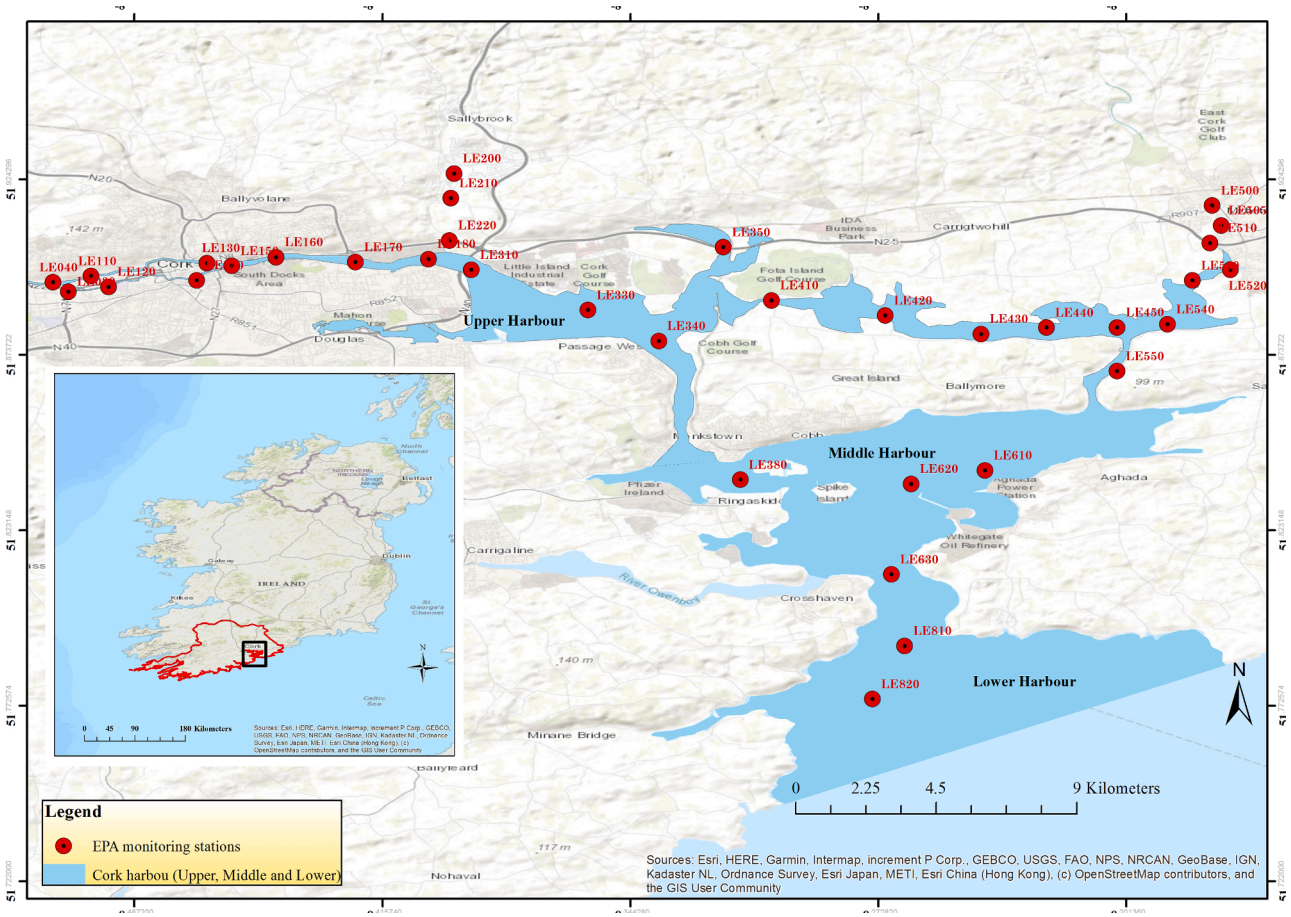


Fig. 2. Cork Harbour and EPA's water quality monitoring sites.

uncertainty. The analysis and findings are presented in section three, and finally in section four conclusions are drawn from this research.

2. Materials and methods

Eight widely used WQI models were used to calculate the WQIs, and the MCS technique was utilized to calculate the uncertainties. Several statistical tools, techniques, and ML approaches were used in this study to evaluate the MCS results. The methodology developed in this paper is presented graphically in Fig. 1 and described in detail in the following subsections.

2.1. The application domain - Cork Harbour, Ireland

The proposed methodology was applied and tested in Cork Harbour, located on the south coast of Ireland (Fig. 2). Cork Harbour was chosen as it was used for a previous WQI model study by the authors (Uddin et al., 2022a) and data were therefore already available. The Harbour, designated as a Special Protection Area (SPA), is Ireland's deepest and longest (17.72 km) surface water body (Hartnett and Nash, 2015; Nash et al., 2011). The area adjacent to the Harbour is one of Ireland's most economically promising zones. Cork Harbour receives large quantities of effluent from various sources each year, such as industries, domestic and municipal waste, which results in variable water quality throughout the harbour (Uddin et al., 2020a,b, 2023).

2.2. Data collection

The Irish Environmental Protection Agency (EPA) collects monthly WQ data from monitoring sites in Irish waterbodies. The relevant water quality indicator data for Cork Harbour were obtained from https://www.catchments.ie/data/. Data for eleven water quality indicators including ammonia (AMN), transparency (TRAN), total organic nitrogen (TON), temperature (TEMP), dissolved oxygen (DOX), ammoniacal nitrogen (AMN), pH, salinity (SAL), molybdate reactive phosphorus (MRP), biological oxygen demand (BOD), and Chlorophyll a (CHL) from 29 monitoring sites were sampled at 1 m depth and at with these data being used in this study. The monitoring locations are presented in Fig. 2. Water quality data from 29 out of 36 monitoring stations in Cork Harbour were obtained from the EPA for the period 2019 – 2020. The detailed water quality data are provided in Table S1 and Table S2 for the summer and winter periods, respectively.

2.3. WQI model

The WQI model of Uddin et al. (2022a) was used for the study as it was developed specifically for coastal waters and was tested using Cork Harbour. Like most WQI models, it involves four main stages: (1) selection of WQ indicators for inclusion, (2) calculation of sub-indices for each indicator using SI functions, (3) calculation of weightings for each indicator based on their influence on water quality, and (4) calculation of the final WQI using aggregation functions. The model was run for summer and winter periods. With regard to parameter selection, following Uddin et al. (2022a), seven water quality indicators were included for summer (CHL, DIN, AMN, TEMP, BOD, MRP and TRAN) and five were included for winter (DIN, TON, pH, DOX and TRAN).

The following SI functions were used to transfer various water quality data onto a uniform scale.

$$SI = (SI_u - SI_l) - \frac{(SI_u \times WQ_m)}{(STD_u - STD_l)} \quad (1)$$

$$SI = \frac{(WQ_{i_m} - STD_l)}{(STD_u - STD_l)} \times SI_u \quad (2)$$

$$SI = (SI_u - SI_l) - \frac{(WQ_{i_m} - STD_l)}{(STD_u - STD_l)} \quad (3)$$

SI_l and SI_u are the lower and upper limits of possible SI values (0 and 100, respectively). STD_l is lower threshold value, STD_u is the upper threshold value and WQ_m is the measured water quality parameter concentration. Parameter weighting was determined using the two-step process of Uddin et al. (2022a) involving XGBoost and the rank order centroid (ROC) methods.

Table S3 and Table S4 provide lists of the water quality indicators included in the study, corresponding weight values and binary conditions for sub-indexes functions; these were all pre-determined by Uddin et al. (2022a). Like Uddin et al. (2022a), eight different aggregation function are tested here; four of these were weighted and four unweighted. Details of the aggregation functions can be found in Annex Table S5. Table S6 provides the SI results obtained from the Eqs. (1) and (2) SI functions and the binary rules (from Table S4) for considered indicators. The outputs (WQI scores) from different aggregation functions are presented in Table S1 and Table S2 for summer and winter, respectively.

2.4. Model uncertainty

The Monte Carlo simulation (MCS) approach was used to quantify uncertainties at each stage of the WQI model application. The MCS technique is a widely recognised method for the quantification of all types of uncertainties in a system/model. Parametric and nonparametric MCS methods were utilised to generate target random samples and then to calculate the uncertainties for each model component. Fig. 1 provides an overview of the experimental procedure and the following section provides a brief description of the MCS method. The four WQI model stages were divided into two categories for uncertainty analysis:

(1) Model input – including the selection, sub-indexing and weighting of water quality indicators, and
(2) Model output – including aggregation of the final indices.

2.4.1. Estimation of uncertainty in WQI model

The GUM methodology was used to implement the MCS method for the estimation of uncertainties in the model. Five common statistical measures of uncertainty were employed; these were: (1) the standard uncertainty (SU), (2) the expanded uncertainty (EU), (3) the combined uncertainty (CU), (4) the coverage factors (CF) and (5) the degree of freedom (df).

(i) Standard uncertainty. The standard uncertainty (SU) is widely used to estimate uncertainty in hydrological/hydraulic models (Liang et al., 2016; Liu et al., 2008; Rodríguez et al., 1989). It refers to the degree of data dispersion from the mean of observations (Farrance and Frenkel, 2012). The method requires the definition of all input entities of the model - here the selected WQ indicators, sub-indexes and weight values. For the model error of an input variable *I*, the SU can be defined for each quantity as follows:

(i) Water quality indicator uncertainties;

$$SU_p = I(WQ_{p_i}) + \alpha_i \quad (12)$$

(ii) Sub-index (SI) uncertainties;

$$SU_{si} = I(SI_i) + \alpha_i \quad (13)$$

(iii) Weight uncertainties,

$$SU_w = I(w_i) + \alpha_i \quad (14)$$

(iv) Model output (aggregation functions) uncertainties;
(a) weighted model:

$$SU_{WQI_w} = WQI(WQ_{p_i}, SI_i, w_i) + \alpha_i \quad (15)$$

(b) unweighted model:

$$SU_{WQI_{uw}} = WQI(WQ_{p_i}, SI_i) + \alpha_i \quad (16)$$

where, SU_p, SU_{si} and SU_w are the water quality indicators, sub-index and parameter weight values, respectively. SU_{WQI_w} is the weighted model output uncertainty and $SU_{WQI_{uw}}$ is the unweighted model output uncertainty. WQ_{p_i} is the ith value of the water quality indicator *I*. SI_i is the sub-index value and w_i is the weight value of ith variable *I*. α is the variation between predicted and observed MCS values. The difference between predicted and observed variances are defined as follows:

Variance,

$$V(x) = V(\alpha_1) + V(\alpha_2) + \ldots + V(\alpha_n) \quad (17)$$

If all of the measured and observed random variations (α) are uncorrelated, then standard uncertainty can be defined as:

$$SU = s_{\bar{x}} = \sqrt{V(\bar{x})} = \frac{sd_{\bar{x}}}{\sqrt{N}} \quad (18)$$

where, $SU_{\bar{x}}$ is the standard uncertainty, $s_{\bar{x}}$ is the standard error of the mean value of each input components, and $sd_{\bar{x}}$ is the mean standard deviation of the random variations in the quantity being measured with the sample true mean. The $s_{\bar{x}}$ can be expressed as:

$$s_{\bar{x}} = \left(\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2\right)^{1/2} \quad (19)$$

where, N is the number of random samples, x_i is the ith data value and $\bar{x}$ is the sample mean.

2.4.2. Combined uncertainty (CU)

In a model with multiple model components, each model component is a source of uncertainty. The combined uncertainty (CU) describes how the model result varies with changes in each model component (Meyer, 2007). In the case of the WQI model, the parameter selection, sub-indexing process and weighting technique all contribute to the input uncertainty. The CU is therefore a measure of the total magnitude of uncertainty associated with all inputs into the WQI model.

The quadrature Root Sum of the Squares (RSS) method is widely used to calculate the CU value in a model (Mishra, 2009). RSS is a measurement of the standard uncertainty for each variable, including the coefficient sensitivity. In this method, all PDFs are combined and converted into the normal distribution based on the central limit theorem (Guimaraes Couto et al., 2013). In this study, RSS was used for quantifying CU as follows:

$$CU = \left[\sum_{i=1}^{n}[c_i SU(x_i)]^2\right]^{1/2} \quad (20)$$

where, CU is the combined uncertainty, c_i is the sensitivity coefficient, and $SU(x_i)$ is the standard uncertainty of the ith water quality indicator. The sensitivity coefficient can be represented as follows:

$$C_i = \frac{\partial f(x_i)}{\partial x_i} = \frac{\partial y_i}{\partial x_i} \quad (21)$$

where, $\frac{\partial f(x_i)}{\partial x_i}$ is the changing value of the inputs of the WQI model (i.e. the model components water quality indicators, sub-indexes and weight values), and $\frac{\partial y_i}{\partial x_i}$ is the aggregation process of the WQI model that are changing by independent model input variables.

2.4.3. Expanded uncertainty (EU)

Expanded uncertainty is another statistical tool that can be used to quantify how much of the measurement's range can be assumed to cover a significant portion of the data distribution (Loucks, P. D., van Beek, 1981). It is anticipated to include a significant portion of the measured value distribution that can be reasonably linked to the model prediction (BIPM et al., 2020). Typically, the EU estimates are used to improve the level of confidence in the uncertainty of measured results. It usually expands the confidence interval by assuming the t-distribution function of the student t- test. The EU is calculated as follows:

$$EU = k \times U_c(y)(p\%) \quad (22)$$

where, y are the random input variables, and k is the coverage factor and U_c combine uncertainty of random data. The coverage factor can be defined as:

$$k = t_{v,1-\alpha/2} \quad (23)$$

where v is the effective number of degree of freedom, $\alpha = 1 - P$, and t is the Student's t test (well known as t-test) variable. In this research, 95% confidence level (P) is used, v is calculated form the Welch–Satterthwaite equation that can be defined as:

$$v = \left(s_{\bar{x}}^2 + \sum_g b_{x,g}^2\right)^2 \Big/ \frac{s_x^4}{v_{s_x}} + \sum_g \frac{b_{x,g}^2}{v_{b_{x,g}}} \quad (24)$$

where, v_{s_x} is the variance of the measured data for N -1 number of degrees of freedom is = 28 (in this study), whereas $v_{b_{x,g}}$ is the variance systematic error (Shaw, 2017) defined as:

$$v_{b_{x,g}} = \frac{1}{2U_{x,\,g}^2} \quad (25)$$

where, $U_{x,g}$ is the relative uncertainty of $b_{x,g}$.

The uncertainty of the WQI model was calculated in a multi-step process as follows:

1. PDFs of all input components were generated.
2. Prior to performing the MCS, the M numbers of trials were set to define measurand and input entities in MCS. The trials M can be represented as follows:

$$M > \frac{10^4}{1-p} \quad (25)$$

where, p is the coverage probability. For $p = 0.95$, the number of trials was $2e^5$. Commonly, higher number of trials is recommended in order to obtain greater convergence in the results.

3. After M trials, the N random values for α were estimated and applied to [equations (12) – (16)] calculate the SU_p, SU_{si} and SU_w, SU_{WQI_w} and $SU_{WQI_{uw}}$ values.
4. For completed M trials, the $SU_n(SU_p,\ SU_{si}$ and $SU_w,\ SU_{WQI_w},\ SU_{WQI_{uw}})$ value calculated in step (3) was used to obtain the statistics of the parameter $s_{\bar{x}}$. Uncertainty can be treated as "stable" if there is a numerical tolerance that corresponds to the standard deviation of the output quantity; this tolerance should be less than two standard deviations of the input quantity. The standard uncertainty can be determined from $\times\ 10^i$, where i is the integer with a number of decimal places. The numerical tolerance can be expresses as:

$$\tau = \frac{1}{2}10^i \quad (26)$$

5. For the optimized MCS algorithm, the steps (3) and (4) are repeatedly run for M trails to generate the s_x value for each trail.
6. Finally, the p% coverage intervals for $s_{\bar{x}}$ values were evaluated, summarized and interpreted in the WQI model for the MCS output.

2.5. Prediction of WQI model uncertainty

2.5.1. Machine learning algorithms

ML technique is widely used for predicting unknown objects or analysis of the model performance (Elgeldawi et al., 2021; Haghiabi et al., 2018; Zhu et al., 2022). Recently, several studies have utilized this technique for estimating the model uncertainty with a 95% CI. In addition, a number of studies have utilized this technique for identifying the faults or biased nature of any model/system (Bode et al., 2020; Ekici, Unal and Ozleyen, 2019; Fernandes et al., 2022; Saufi et al., 2019; Yang et al., 2019). A few studies have utilized this technique for estimating water quality model uncertainty (Jiang et al., 2021; Khoi et al., 2022; Xu et al., 2022). In this study, the ML technique was utilized to quantify uncertainty in the WQI model because it focuses mostly on population variables that determine the prediction intervals. The calculations of uncertainty of each specific prediction can be fairly expensive computationally and usually requires an additional computational effort beyond the training of the model (Abdar et al., 2021; Tavazza et al., 2021).

2.5.1.1. ML models development processes

2.5.1.1.1. Data preparation.

(a) Data standardization

Prior to develop the ML models, it is important to standardize the data for avoiding the model over-fitting problems (Rahman, 2019). Usually, in ML technique, this approach used to convert various types of variables into uniform scale. In this research, data was standardized using the approach of Uddin et al. (2022c). Details of the methodology can be found in Uddin et al. (2022c).

(b) Preparing training and testing datasets

For the purposes of developing the ML models, dataset was divided into two sets (i) 70 % (20) of training and 30% (9) of testing. The detail input datasets are provided in Table S1 and Table S2 for summer and winter, respectively.

2.5.1.1.2. Development of ML models. After preparing datasets for training and testing, six ML-regression algorithms including (1) decision tree (DTs), (2) linear regression (LR) model, (3) Gaussian process regression (GPR) model, (4) support vector machines (SVMs), (5) neural network (NN), and (6) ensembles of tree model were trained and tested using training (20 monitoring sites) and testing (9 monitoring sites), respectively, whereas the model performance were evaluated during both training and testing phases. Details of the ML algorithms can be found in Uddin et al. (2022c).

The best-fit ML algorithm was selected from six ML techniques based on the trial-and-error analysis using the regression learner app with MATLAB R2021b (Bui et al., 2020). For obtaining the best-fit ML algorithm(s), many hyperparameter tuning techniques are used in ML

Table 1
Optimized value for GPR model.

Model attributes	Optimized value	Hyperparameter search range
Signal standard deviation	10.8369	-
Basis function	Constant	Constant, Zero, Linear
Kernel function	Nonisotropic Squared Exponential	Nonisotropic Exponential, Nonisotropic Matern 3/2, Nonisotropic Matern 5/2, Nonisotropic Rational Quadratic, Nonisotropic Squared Exponential, Isotropic Exponential, Isotropic Matern 3/2, Isotropic Matern 5/2, Isotropic Rational Quadratic, Isotropic Squared Exponential
Kernel scale	0.25294	0.009433-9.433
Sigma	151.5629	0.0001-153.2569
Standardiz	false	true, false
Iterations	30	-

studies, the grid search technique is one of them (Elgeldawi et al., 2021; Shekar and Dagnew, 2019; Uddin et al., 2022c). In this research, the grid search technique was utilized according to the methodology of Shekar & Dagnew, (2019) because, compared to other methods, this technique is effective in optimizing the model parameters. The present study was utilized the regression learner app with MATLAB R2021b for the identification of optimal parameters of ML models using auto tuning approach.

Based on the trial and error performance results of six ML models, the GPR ML model was found to produce the minimum training error and best fit for predicting WQI input dataset. Details of the GPR-ML algorithm can be found in Grbić et al. (2013). Table 1 provides the optimal parameters for the best-fitted GPR-ML in predicting various WQI model scores. The GPR, which incorporates Bayesian theory and statistical learning theory, is an effective method for solving complex regression problems such as high dimensionality and nonlinearity (Zhang et al., 2019). In terms of uncertainty analysis, a few studies have reveal that the GPR is different from traditional ML methods in that it automatically determines the prediction uncertainty when the model is fit (Rasmussen, 2004). Abdar et al., 2021 suggest that the GPR algorithm has a

Table 2
Assessment of water quality in in Cork Harbour at monitoring site using different WQI models.

WQI methods	water quality classifications summer				winter			
	Good	Fair	Marginal	Poor	Good	Fair	Marginal	Poor
(i) Weighted								
NSF	45% (13)	24% (7)	31% (9)	0	14% (4)	24% (7)	17% (5)	45% (13)
WQM	62% (18)	38% (11)	0	0	14% (4)	48% (14)	38% (11)	0
SRDD	28% (8)	34% (10)	7% (2)	31% (9)	10% (3)	10% (3)	5	17
WJ	41.5% (12)	17% (5)	0	41.5% (12)	14% (4)	14% (4)	10% (3)	18
(ii) Unweighted								
RMS	15	48% (14)	0	0	24% (7)	22	0	0
AM	9	20	0	0	6	41.5% (12)	11	0
Hanh	14% (4)	24	0	0	17% (5)	28% (8)	10% (3)	45% (13)
CCME	100% (29)	0	0	0	100% (29)	0	0	0

In total 29 monitoring sites were used in this assessment.

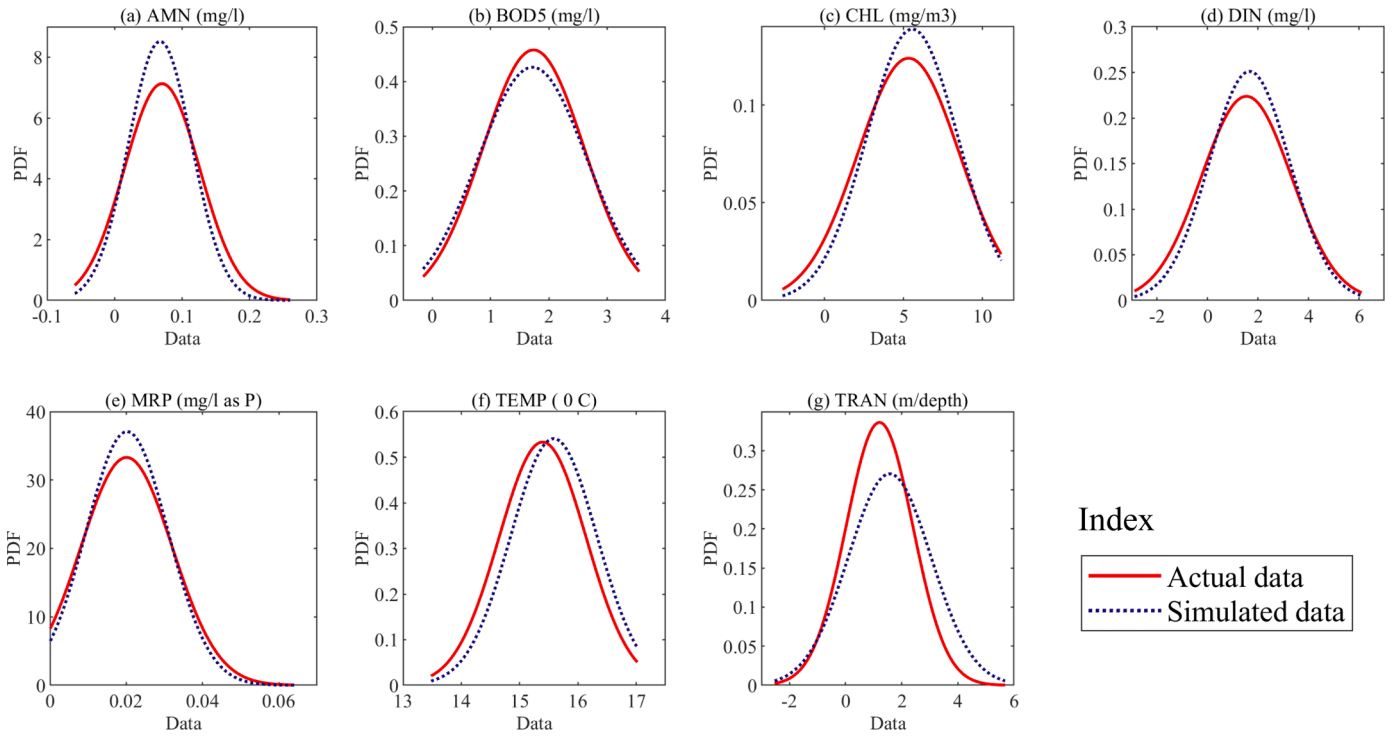


Fig. 3. PDF plots of actual and simulated water quality indicators data in Cork Harbour for the summer period.

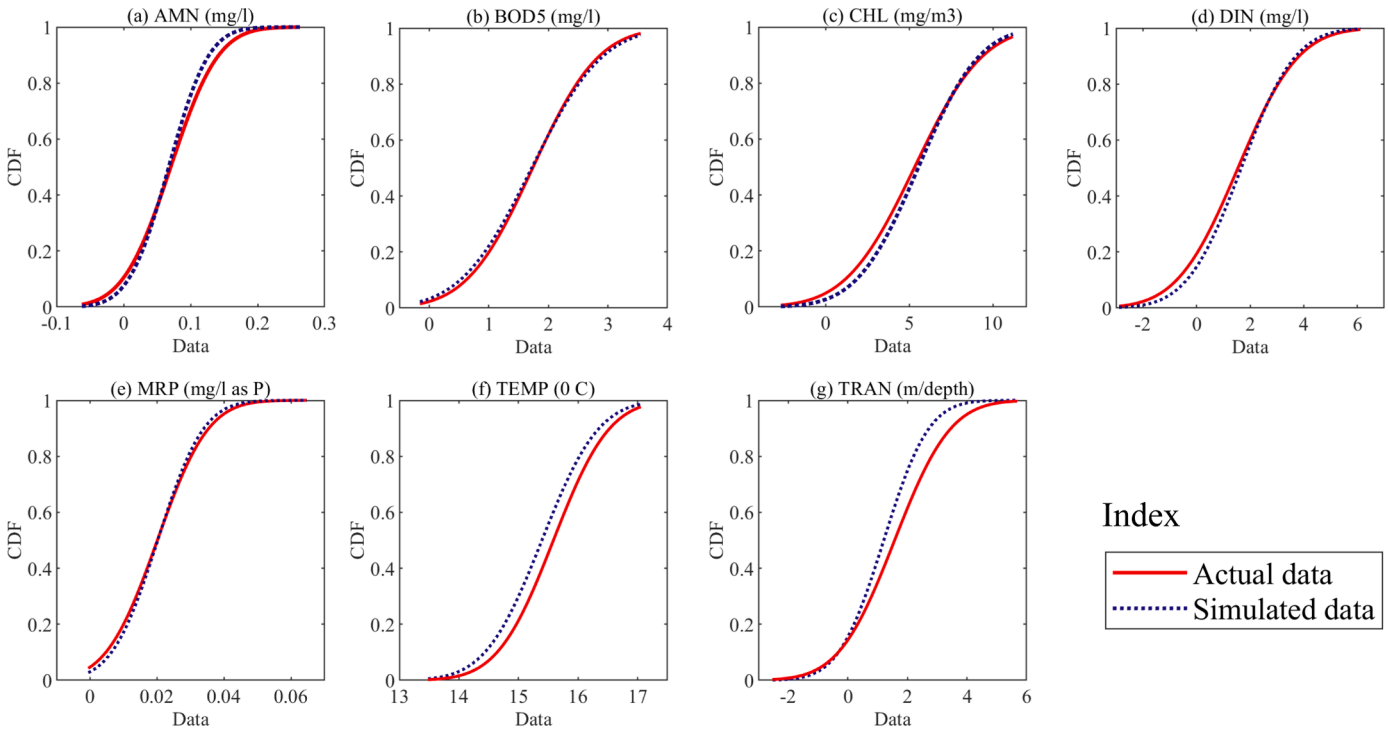


Fig. 4. CDF plots of actual and simulated water quality indicators data in Cork Harbour for the summer period.

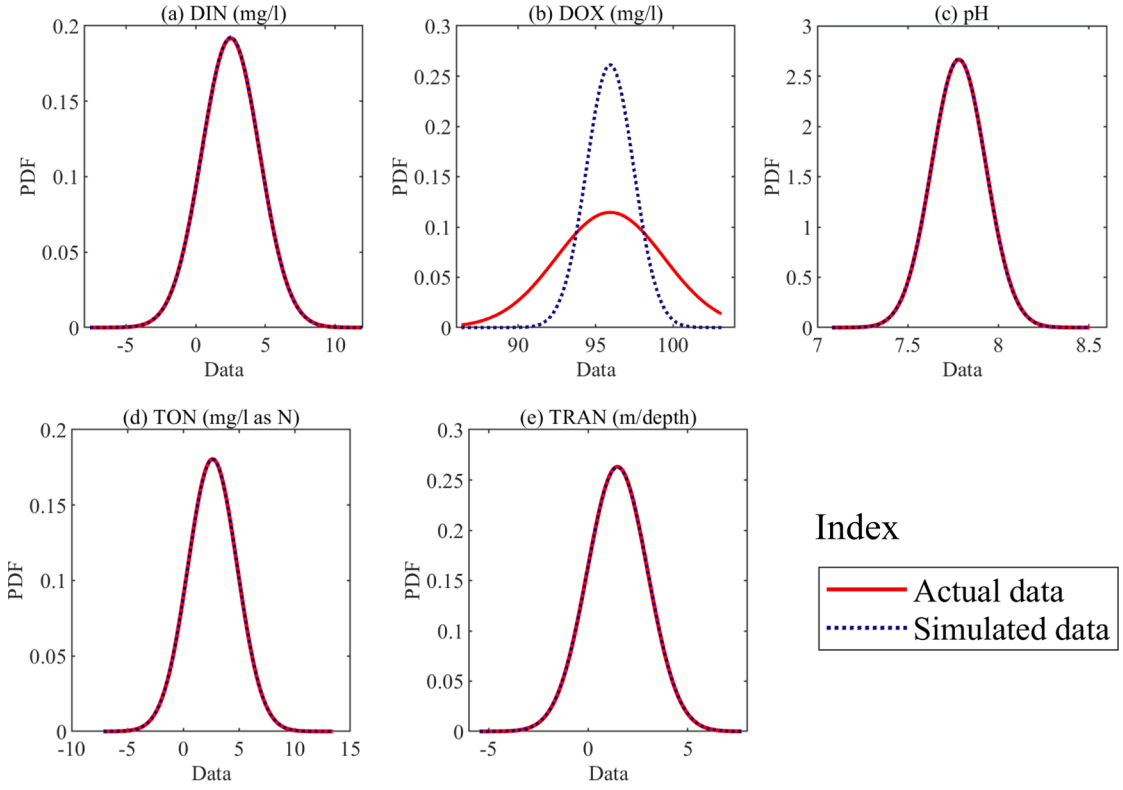


Fig. 5. PDF plots of actual and simulated water quality indicators data in Cork Harbour over the winter season.

higher predictive accuracy than other algorithms and the technique has been successfully applied for predicting model uncertainty (i.e. Asadollah et al., 2021; Tavazza et al., 2021). Therefore, after obtaining the robust algorithm, the GPR-ML model was used for predicting WQI values with 95% of CI in order to estimate the model uncertainty. Once the WQI values and confidence intervals (CI) were predicted, the uncertainty was modelled using the inferential error (Cumming et al., 2007; Rahman and Harding, 2017):

$$Inferential\ CI = M\ \pm t_{(n-1)} \times SE \tag{27}$$

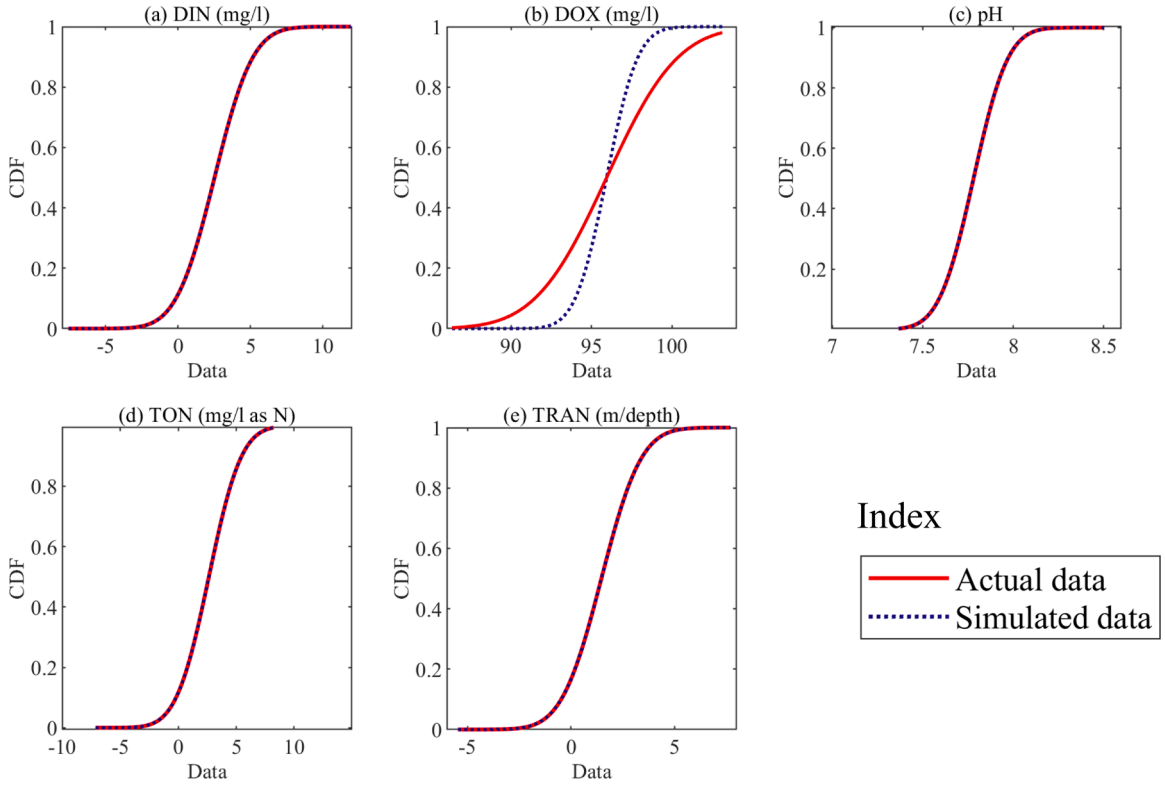


Fig. 6. CDF plots of actual and simulated water quality indicators data in Cork Harbour over the winter period.

Table 3
Estimating uncertainty for model inputs with 28 effective degree of freedom, 95% coverage probability and 2.05 coverage factor.

Input entities		summer		winter		
WQ paramters	SU[1]	EU[2]	CU[3]	SU	EU	CU
CHL	0.599	1.227				
AMN	0.011	0.021				
TEMP	0.137	0.281	***0.916***			
BOD	0.162	0.331				
MRP	0.002	0.004				
TRAN	0.268	0.549		0.282	0.577	
DIN	0.331	0.678		0.384	0.788	
TON				0.41	0.842	***0.812***
DOX				0.283	0.581	
pH				0.027	0.057	
Sub-indexing						
CHL	4.423	9.059				
AMN	5.34	10.94				
TEMP	0	0				
BOD	2.985	6.115	***12.836***			
MRP	3.819	7.823				
TRAN	7.076	14.49		7.412	15.18	
DIN	4.495	9.208		3.674	7.526	
TON				4.571	9.363	***10.27***
DOX				1.005	2.059	
pH				0.097	0.199	
Weighting	0.0228	0.0467	***0.183***	0.027	0.056	***0.205***
Combined input uncertainity			**13.936**			**11.29**

[1] Standard uncertainty
[2] Expanded uncertainty
[3] Combine Uncertainty

where, M is the observation mean, $t_{(n-1)}$ is a critical value of t. SE is the standard error of observations.

2.5.2. Performance metrics

The cross validation approach is widely used to evaluate the ML model performance for relatively small dataset (Rahman et al., 2013; Xiong et al., 2020). Following Uddin et al. (2022c), the 10-fold cross validation technique was utilized to evaluate prediction performance of the GPR model. The prediction performance was assessed using the standard error metrics: root mean square error (RMSE), mean absolute error (MAE), mean square error (MSE) and coefficient of determination (R^2), defined mathematically as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{\mathrm{n}}\sum_{\mathrm{i}=1}^{\mathrm{n}}(\mathrm{y_i} - \widehat{\mathrm{y}}_\mathrm{i})^2} \tag{28}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y_i}| \tag{29}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2 \tag{30}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y_i})^2}{\sum_{i=1}^{n}(y_i - \widehat{y})^2} \tag{31}$$

where y_i is the actual value of WQI, whereas$\widehat{y}$ is the mean predicted WQI value for the i[th] monitoring site; n is the number of water quality monitoring sites.

2.5.3. Model differences

Tukey's (reference) honestly significant difference post-hoc test (HSD) is a widely used statistical approach for differentiating the pair-wise comparison among models/methods/datasets and was used here to compare the pair-wise WQI models (Goeman and Solari, 2021). The

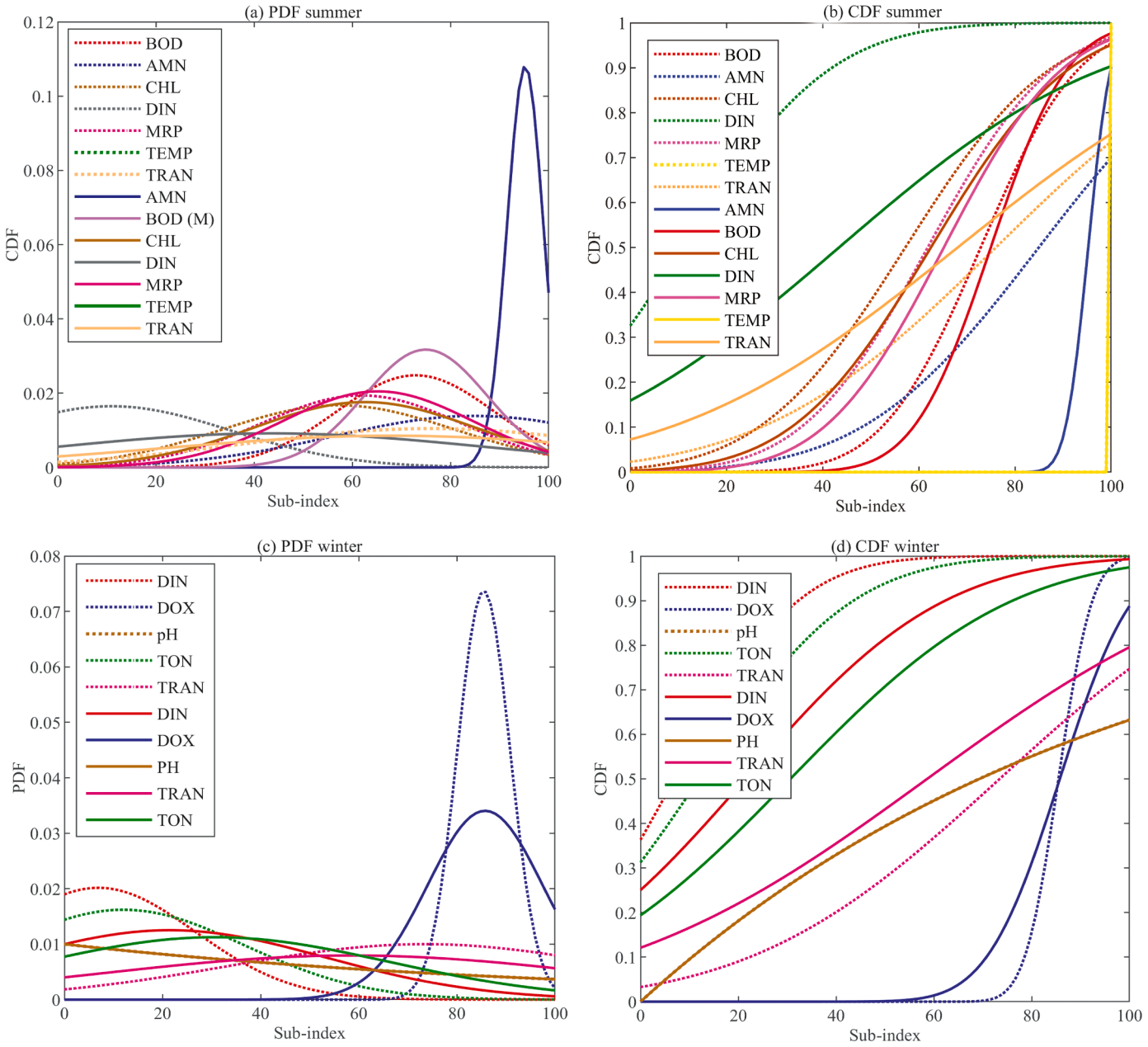


Fig. 7. PDF and CDF plots of calculated and simulated sub-index values for (a,b) summer and (c,d) winter. Dashed lines are calculated SI values and solid lines are simulated values.

analysis of variance (ANOVA) using Welch's test was applied to validate the significance differences between WQI models. Recently, a number of studies have utilized this approach to compare the overall differences between outputs of models (Goeman and Solari, 2021; Rouder et al., 2016; Wang et al., 2017). HSD for each pair of mean values was estimated as follows

$$HSD = \overline{X}_i - \overline{X}_j \mp \frac{q}{\sqrt{2}} \sqrt{MS_e \left(\frac{1}{N_i} + \frac{1}{N_j}\right)} \tag{32}$$

where $i \neq j$:, $\frac{q}{\sqrt{2}} \sqrt{MSE \left(\frac{1}{N_i} + \frac{1}{N_j}\right)}$ is the margin of an error for the confidence intervals, q is the multiplier of the distribution, N is the number of samples within the group, *MSE* is the mean squared errors and, $\overline{X}_i$ and $\overline{X}_j$ are mean values for each group, respectively.

3. Results and discussion

3.1. Comparative assessment of water quality status

The ultimate aim of the WQI model is to assess the status of water quality using a classification scheme. The present study used the classification scheme of (Uddin et al., 2022a). Table S7 contains more information about the classification system. Table 2 provides a comparison of the status of water quality using various WQI models. Tables S8 and S9, for the summer and winter seasons, respectively, show the overall water quality status at each monitoring site in Cork Harbour. From Table 2, it can see that relatively "good" water quality status was found in the summer season compared to the winter period. A significant difference was found in assessing water quality status for various WQI models. The results of WQI models indicate that the assessment results varied significantly due to the model architecture. Similar findings are reported to those of previous studies for Cork Harbour (Uddin et al.,

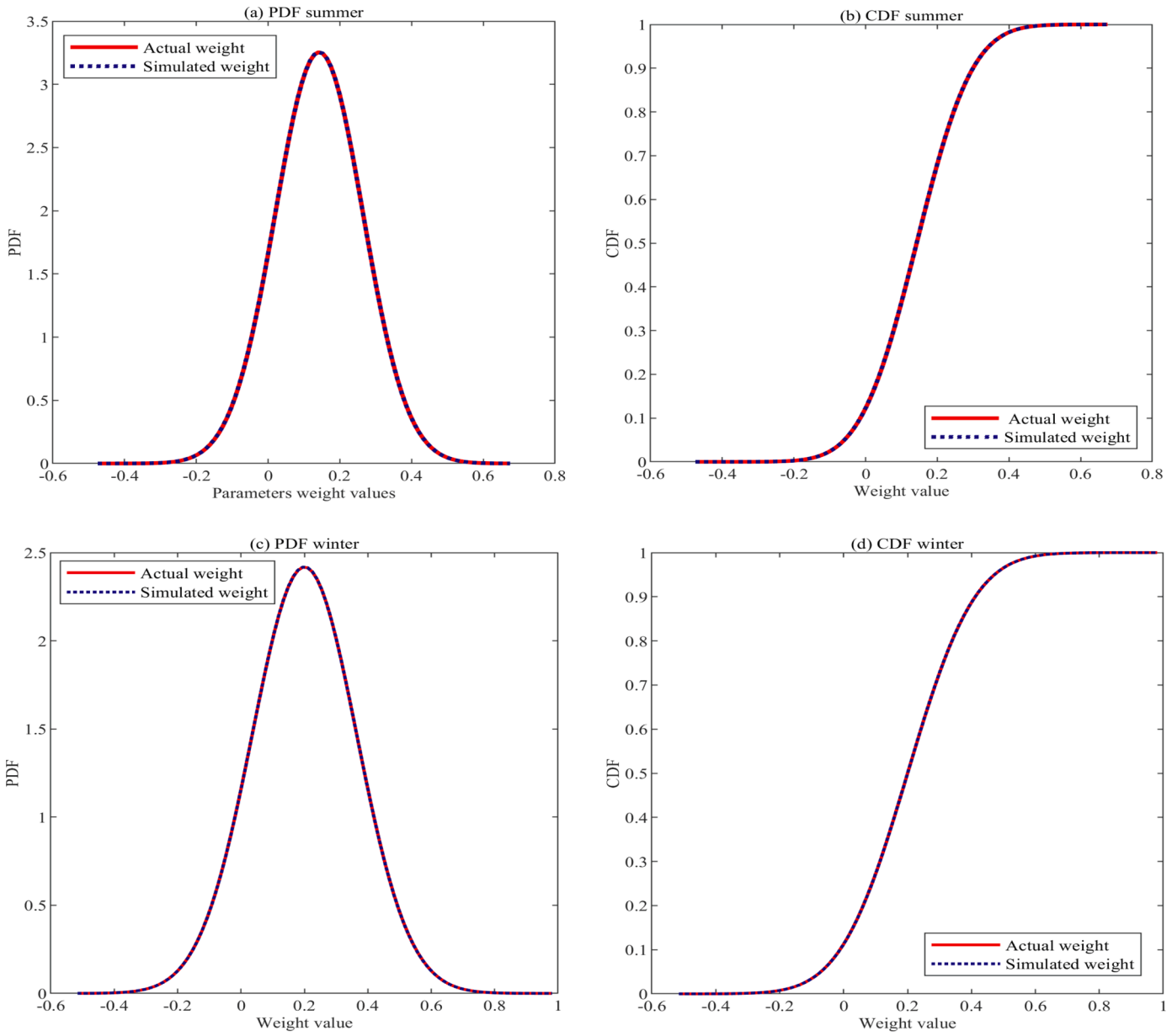


Fig. 8. PDF and CDF plots of estimated and simulated indicators weight values with 95% confidence interval of the probability.

2020a; 2020b; 2022a; 2022c; 2023).

3.2. Initialization of MCS run

The MCS run was initialized for 95% confidence level and 28 degree of freedom including coverage factor k = 2.05 (Meyer, 2007). The probability density function (PDF) of each input parameter is the essential requirement for MCS as they strongly influences the simulation results (Seifi et al., 2020). All input and output PDFs were verified using Kolmogorov-Smirnov test with p-value > 0.05. The MCS simulations were for $M = 2 \times e^5$ trials. The parametric and non-parametric bootstrap tests were carried out to verify the estimated PDFs statistics (Shaw, 2017).

3.3. Sources of uncertainty in WQI model

3.3.1. Model input uncertainty

(i) Water quality indicators uncertainty. The results of the WQ indicators were evaluated graphically and statistically to quantify the input uncertainty of the WQI model. Figs. (3–6) present the actual and MCS-simulated PDFs and cumulative distribution function (CDFs) of all WQI model input entities. All of the WQ indicators were found to be normally distributed during both seasons (Fig. 3 and Fig. 5). The uncertainties calculated for each water quality indicator are presented in Table 3. The PDF statistics and their corresponding confidence intervals for the MCS results are given in Table S10.

From Figs. 3 and 4, there was significant differences between the PDFs and CDFs of the actual and simulated concentrations of water quality indicators during the summer season. Large data variation was observed for CHL, DIN, TEMP, BOD, and TRAN whereas data consistency was found between actual and simulated for AMN and MRP. During winter (Fig. 5–6), there was no significant difference found between the actual and simulated data at $p < 0.0001$, except for DOX.

It can be concluded from the results of the CDFs and PDFs that five water quality indicators (CHL, BOD, DIN, TEMP, and TRAN) contribute to the generation of a small amount of uncertainty in the WQI model through the summer season. From Table 3, less than 1% of uncertainty

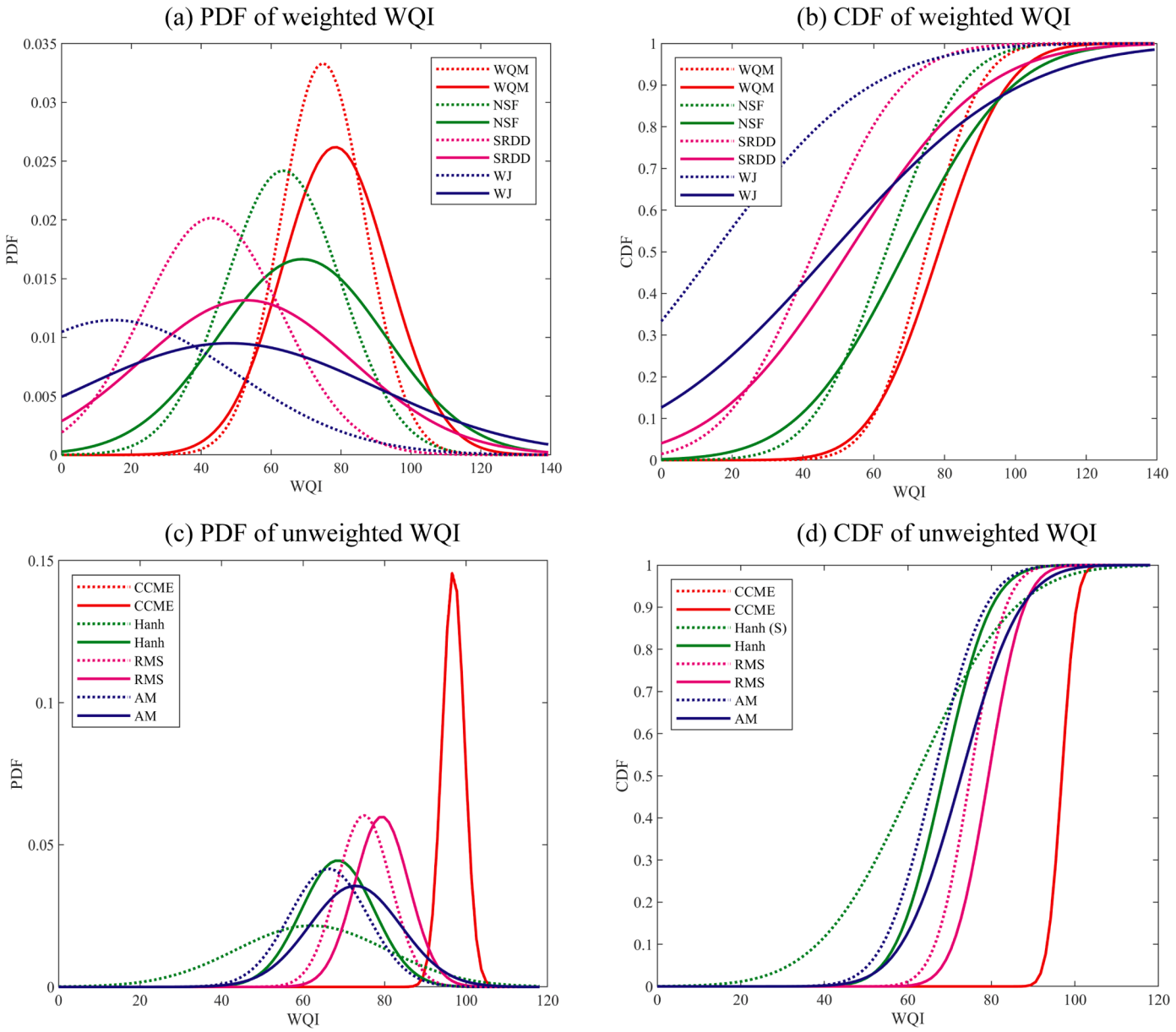


Fig. 9. PDF and CDF plots of calculated and simulated WQI values for different aggregation functions of WQI models over the summer season in Cork Harbour, including 95% confidence level of the probability functions; solid line represents the calculated WQI values and dotted line illustrated for the simulated WQI values.

was associated with all input indicators during both seasons. Seasonal intercomparison shows generally lower uncertainty for the winter period. The results of water quality indicator inputs indicate that the number of input variables plays a crucial role in producing the uncertainties in the WQI model.

(ii) Sub-index uncertainty. Several studies argue that the sub-indexing process generates high levels of uncertainty in a WQI model, but the magnitudes of SI uncertainties have never been quantified. The MCS technique was used to calculate the levels of uncertainty generated during the sub-indexing process. The PDFs and CDFs of SI values are presented in Fig. 7; Table S10 provides the summary of MCS statistical results for distribution functions of sub-indexes. The associated uncertainties for each water quality indicator are presented in Table 3.The PDFs of sub-index functions for all indicators exhibit normal distribution for both seasons. During summer season, there was a significant difference found between the actual and simulated SI values for three parameters: DIN, AMN and TRAN (Fig. 7a,b). Similar differences in SI were observed in winter season for four parameters: DOX, DIN, TON, and TRAN (Fig. 7c,d). There was a strong correlation found between the actual and simulated PDF and CDF of SI values for BOD, MRP, and CHL during summer and for DOX during winter. From Table 3, for summer, 7.07%, 5.34%, 4.49%, 4.42%, 3.81%, and 2.98% of SU were associated with the SI process for TRAN, AMN, DIN, CHL, MRP, and BOD, respectively. Interestingly there was no uncertainty produced by SI of TEMP. For winter, the SI of TRAN produced the highest SU values, whereas pH resulted in the lowest (Table 3). Overall, the combined uncertainty values associated with the SI functions of the water quality indicators were 12.84% and 10.27% for summer and winter, respectively; thus, SI functions play an important role in generating uncertainty in the WQI model during both seasons. It can be concluded that the SI uncertainty varies depending on the spatiotemporal distribution of WQI model inputs.

(iii) Parameter weighting uncertainty. Several studies argue that the weighting process generates a considerable amount of uncertainty that

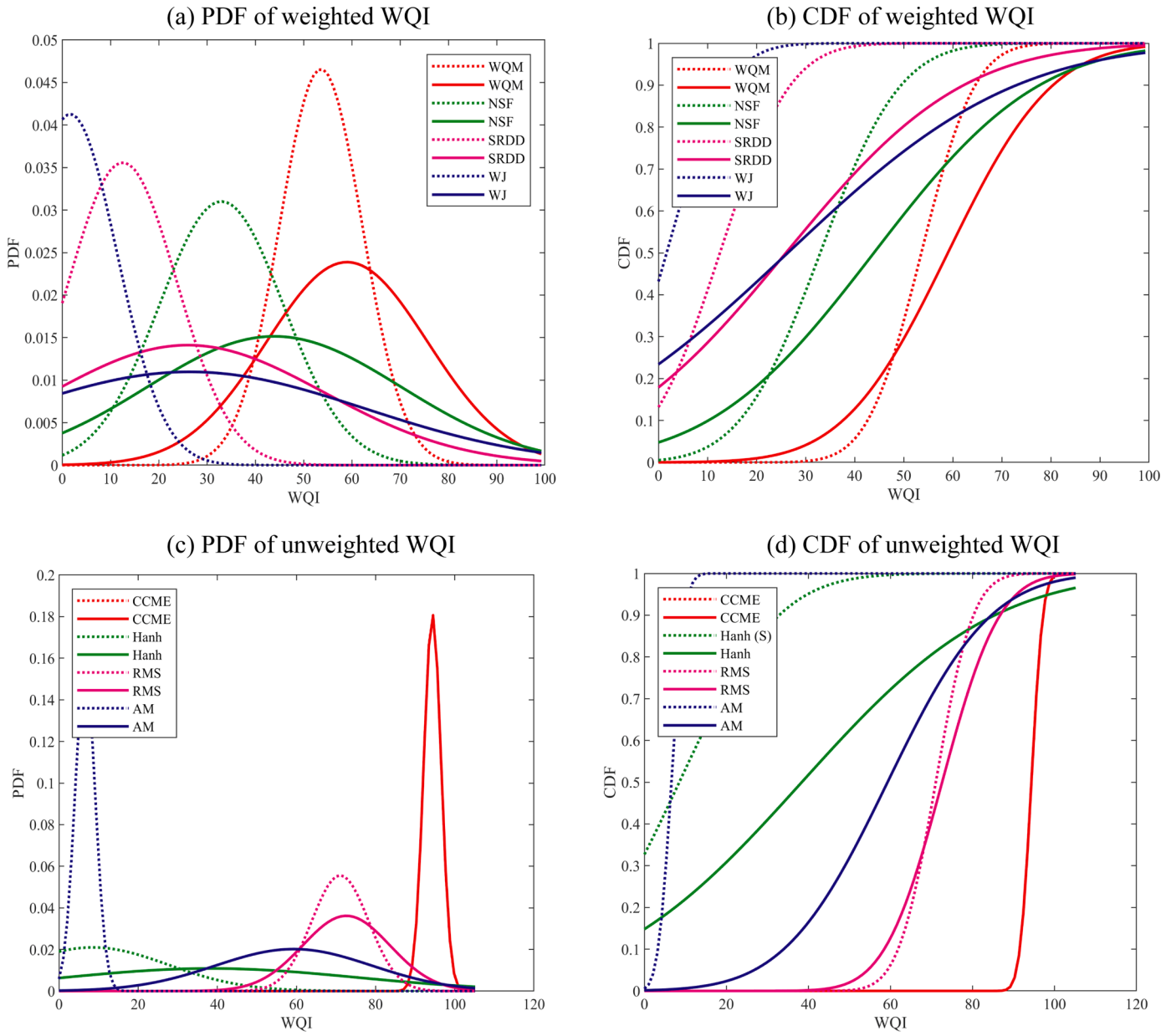


Fig. 10. PDF and CDF plots of calculated and simulated WQI values for different aggregation functions of WQI models over the winter season in Cork Harbour, including 95% confidence level of the probability functions; where solid line represents the calculated WQI values and dotted line illustrated for the simulated WQI values.

contributes to the final WQI score (Seifi et al., 2020; Sutadian et al., 2017, 2016; Uddin et al., 2021). Once again, the MCS approach was utilized to estimate the uncertainties generated in the weighting process. Fig. 8 presents the PDFs and CDFs of the actual and simulated weight values for summer and winter. Table 3 provides the uncertainty results obtained from the MCS analysis of the weight values. Detailed PDFs statistics and uncertainty intervals are presented in Table S10.

The indicator weight values were found to have a normal distribution for both seasons. No weight variations were found between the actual and simulated values for either season. It can be seen from Table 3 that less than 1% of uncertainty is attributed to the weighting process. These findings suggest that the weighting process of Uddin et al. (2022b) does not significantly contribute to uncertainty generation in the WQI model. This is in line with the findings of Uddin et al. (2022a). Unlike the findings of this study, a few studies have revealed that the weight values can contribute a considerable amount of uncertainty in a WQI model (Sutadian et al.,2018).

(a) Impact assessment of input components on the WQI model. Table 3 provides the uncertainty results obtained from the MCS analysis for each input component of the WQI model. Comparing the two seasons, it can be seen that during summer the uncertainty levels of 0.92%, 12.82% and 0.18% were attributed to the water quality indicators, sub-index functions and parameter weight values, respectively. The total uncertainty of the three inputs together is 13.94% (at 95% confidence level) for summer season. For winter, the uncertainties of 0.81%, 10.27%, and 0.21% were attributed to the three processes with a total uncertainty level of 11.29%, which is marginaly lower than the summer score. As for summer, the sub-index function uncertainty level has the most significant impact on the WQI score. Interestingly, and perhaps surprisingly, this study demonstrates that water quality indicators and weight values have no significant impacts on the model score. To our knowledge this is the first research study that quantifies uncertainties at each step of WQI modelling.

(b) Model output uncertainty. The aggregation process is the final step

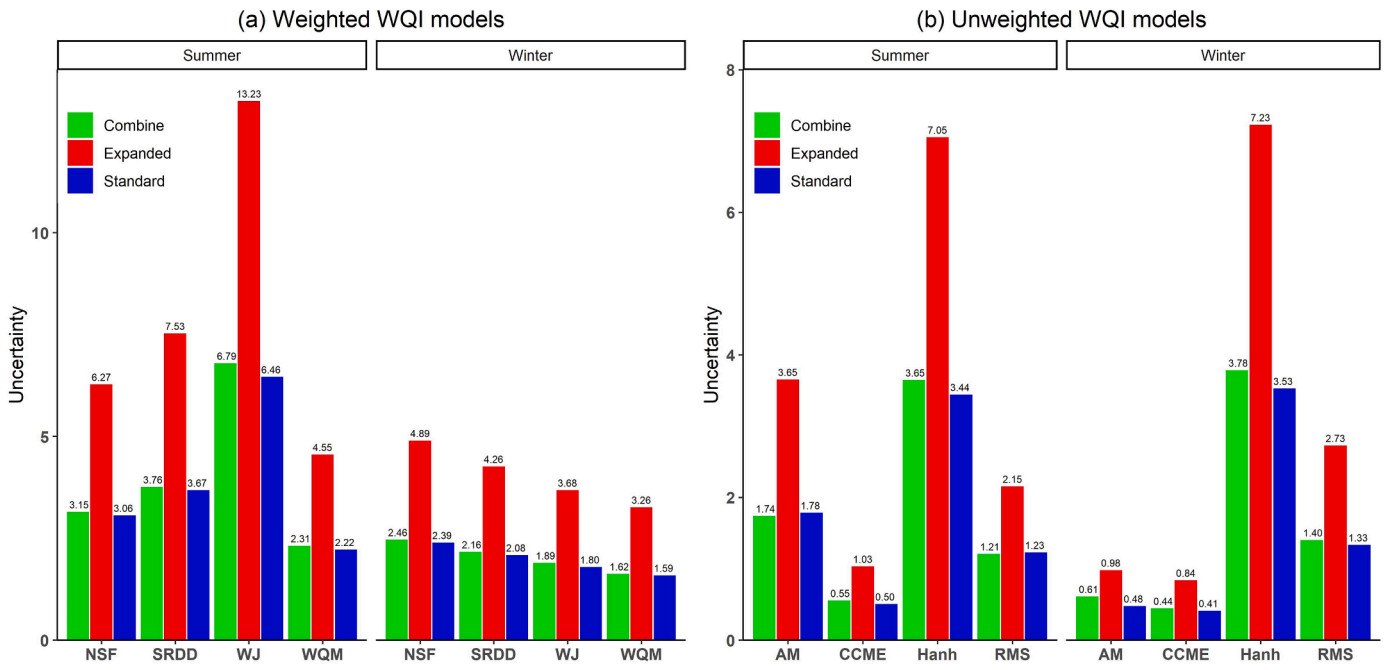


Fig. 11. Uncertainty of different WQI models with 95% coverage probability and 2.05 coverage factor.

Table 4
10-fold cross validation results of GPR prediction model.

WQI Models	Summer			Winter		
	RMSE	MSE	MAE	RMSE	MSE	MAE
NSF	6.018	36.022	4.296	0.390	0.152	0.342
WQM	**4.094**	**16.763**	**2.855**	**0.266**	**0.071**	**0.183**
SRDD	7.622	58.087	6.041	1.296	1.679	0.594
WJ	16.010	373.270	14.508	9.709	94.274	4.462
CCME	**1.331**	**1.772**	**0.967**	**0.643**	**0.458**	**0.440**
Hanh	3.067	9.404	2.465	6.448	41.580	2.224
AM	3.623	13.132	2.794	0.276	0.074	0.247
RMS	**1.887**	**3.561**	**1.526**	**0.532**	**0.282**	**0.306**

*Bold values indicates the higher accuracy of prediction model.

of the WQI model. It is used to aggregate the sub-indices into a single overall WQI score. Eight different aggregation functions were assessed to quantify the uncertainties that they contributed to the WQI score.

The PDFs and CDFs for calculated and simulated results of the eight different aggregation functions (4 weighted and 4 unweighted) are presented in Figs. 9 and 10, respectively. The probability distribution of all aggregation functions were normal during both seasons. Table S11 provides the detailed statistics of MCS. For weighted functions, Significant differences were found between the calculated and simulated WQI values for the SRDD and WJ models over the study period while lower differences were observed between the calculated and simulated WQI values for the NSF and WQM models during both seasons.

For unweighted functions, in general there were no significant data variation between the calculated and simulated WQIs ($p < 0.0001$). The Hanh model exhibits slightly higher differences than the AM and RMS models, but the CCME model shows very small differences (Fig. 9c, 10c, d). The performance of the CCME function could be attributed to its architecture, which differs significantly from the others. Lumb et al. (2011) and Uddin et al. (2022a) found that the CCME model has eclipsing issues linked to overestimation.

Fig. 11 presents the results of the standard, expanded, and combined uncertainty produced by the eight aggregation functions. For the weighted methods, the WJ and the SRDD functions contributed the highest uncertainty to final WQI scores during both seasons while the WQM model contributed the lowest level of uncertainty. During summer, the highest and lowest SU values were produced by the WJ and the WQM models, respectively (6.46 % versus 2.22%). By comparison, during winter, the highest and lowest SU were produced by the NSF and WQM models, respectively (2.39 % and 1.59%) (Fig. 11a). The WJ model exhibits the highest expanded uncertainty range, while the WQM model had the lowest. The expanded uncertainty results indicate that the WJ model could generate a larger data variation than other models. The WJ model exhibits a particularly high sensitivity to inputs during the summer season. This implies that the variations in model inputs result in a significant variation in model outputs. A well balanced combined uncertainty in the input variables of 2.31% and 1.62% were found for the WQM model through the summer and winter, respectively. For the unweighted models, the CCME and Hanh models recorded the lowest and highest levels of uncertainty during both seasons, respectively (Fig. 11b). Similar results for the CCME model were reported in literature (Ahmed et al., 2021; Davies, 2006; Gupta et al., 2003; Lumb et al., 2011; Wu et al., 2021; Zotou et al., 2019)

There is a significant seasonal difference in the uncertainties between the models with generally lower uncertainty levels observed during the winter season. This could be attributed to a variety of reasons, including the number of lower number of WQ variables used in winter, ambiguity of the sub-index functions, sample sizes, and the eclipsing problem of the model aggregation process (Gupta et al., 2003; Seifi et al., 2020; Sutadian et al., 2018; Uddin et al., 2021; 2022a). A comparison of uncertainty levels generated by the weighted and unweighted models shows that the unweighted models produce lower uncertainties. Ignoring the CCME model because of its aforementioned problems, the weighted WQM and unweighted RMS models are recommended for use because they produced the lowest levels of uncertainty. The results of the MCS are in line with those of earlier studies (Carsten Von Der Ohe et al., 2007; Gupta et al., 2003; Lumb et al., 2011; Mladenović-Ranisavljević and Žerajić, 2018; Wu et al., 2021; Zotou et al., 2019).

3.4. *Predicting Uncertainty*

The GPR model was utilized for predicting WQI model uncertainty. Table 4 provides the 10-fold cross validation results of the GPR prediction model for the eight WQI models and for summer and winter. For

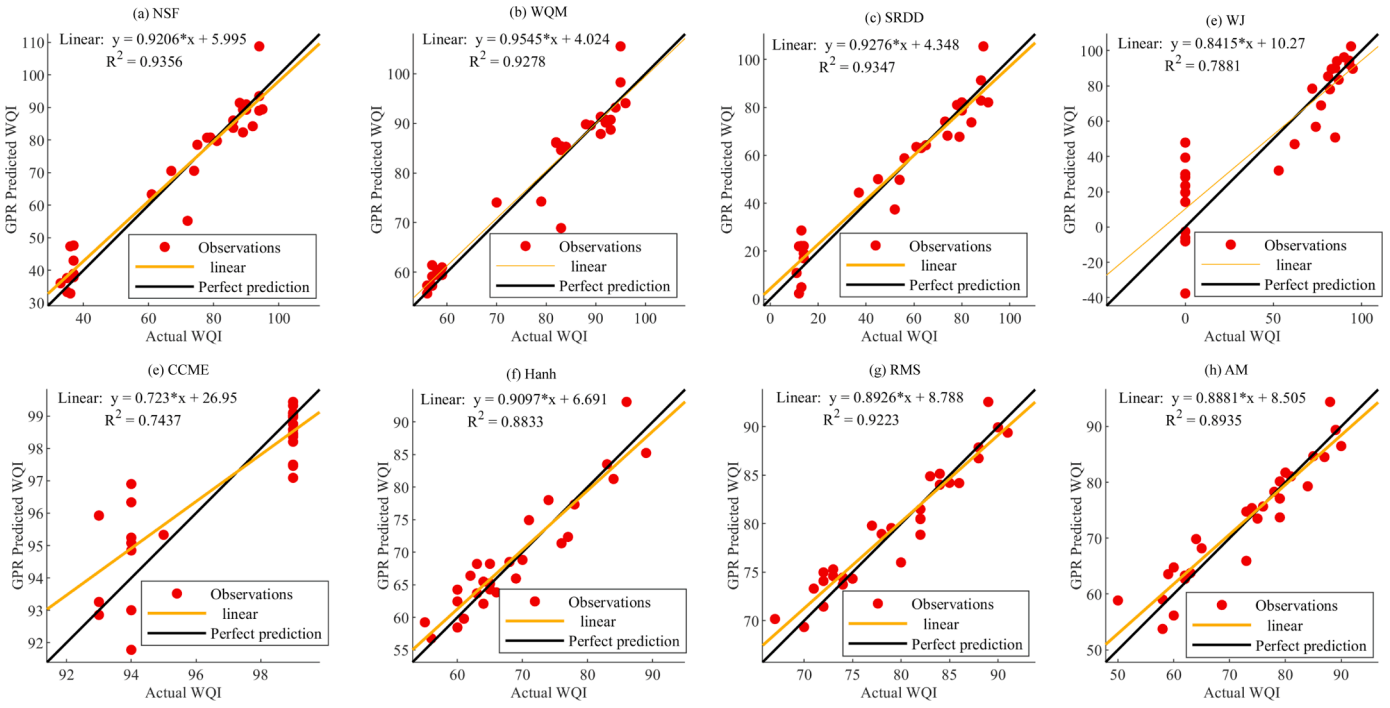


Fig. 12. Scatter plots of GPR predicted and actual (calculated) WQI values of different WQI models for the summer season.

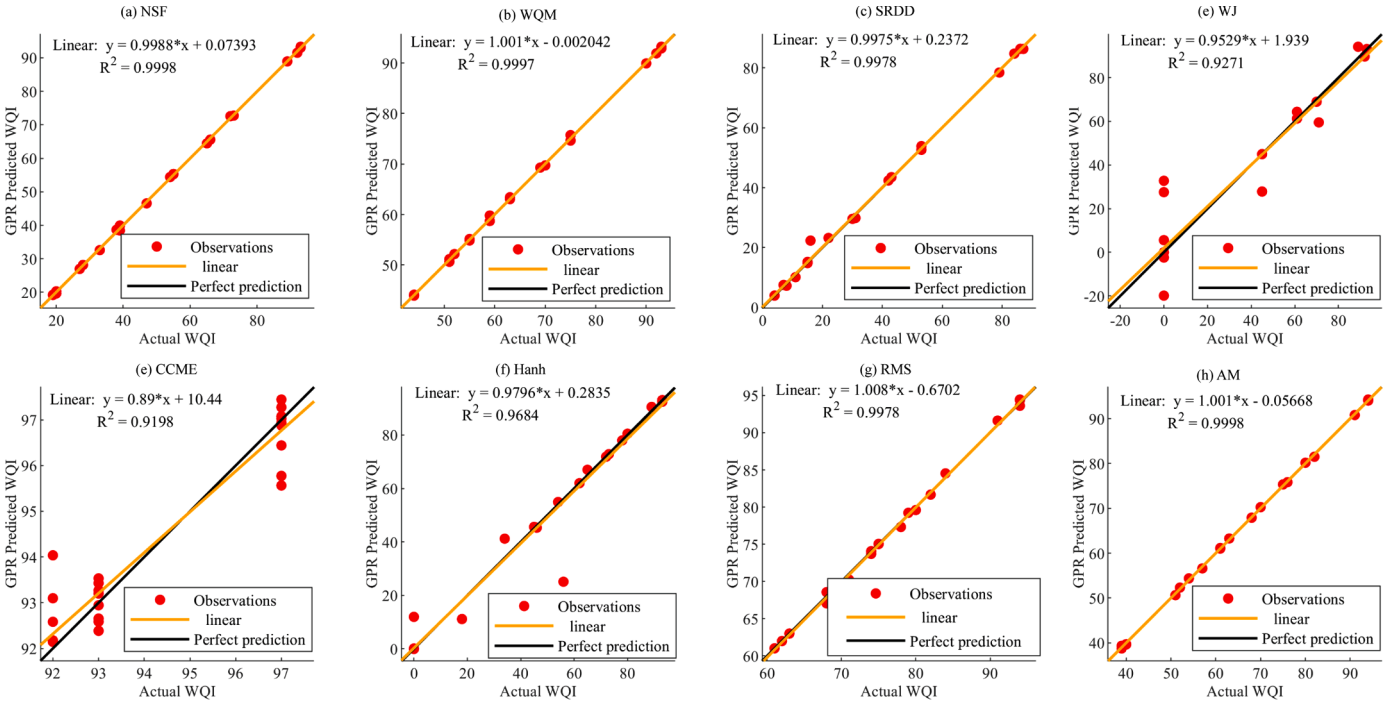


Fig. 13. Scatter plots of GPR predicted and actual (calculated) WQI values of different WQI models for the summer season.

the weighted models, the lowest prediction error was produced for the WQM model, while for the unweighted models the lowest error was for the CCME and RMS models. Fig. 12 and Fig. 13 present linear regression analyses of the GPR predicted scores and those calculated (actual) using the WQI models. Generally there is a good fit of the GPR model for both seasons. For the purposes of assessing the predictors' effect on the model response, the present study utilized the coefficient of determination according to the approach of Uddin et al. (2022a). For the determination of R^2, the present study used twenty-nine samples of water quality data. The coefficient of determination was found to be less than 0.8 was found only for the WJ and the CCME models during the summer season.

Figs. 14 and 15 show the GPR-predicted WQI values, SE and 95% CI for each sampling location in Cork Harbour for summer and winter, respectively. Table 5 provides the results obtained from the t-test analysis of GPR predicted WQI values for the different WQI aggregation functions. The t-test results indicate that the WQI values predicted for the WQM are noticeably different from the calculated values, when compared to the other weighted functions. In addition, lower SE and SD

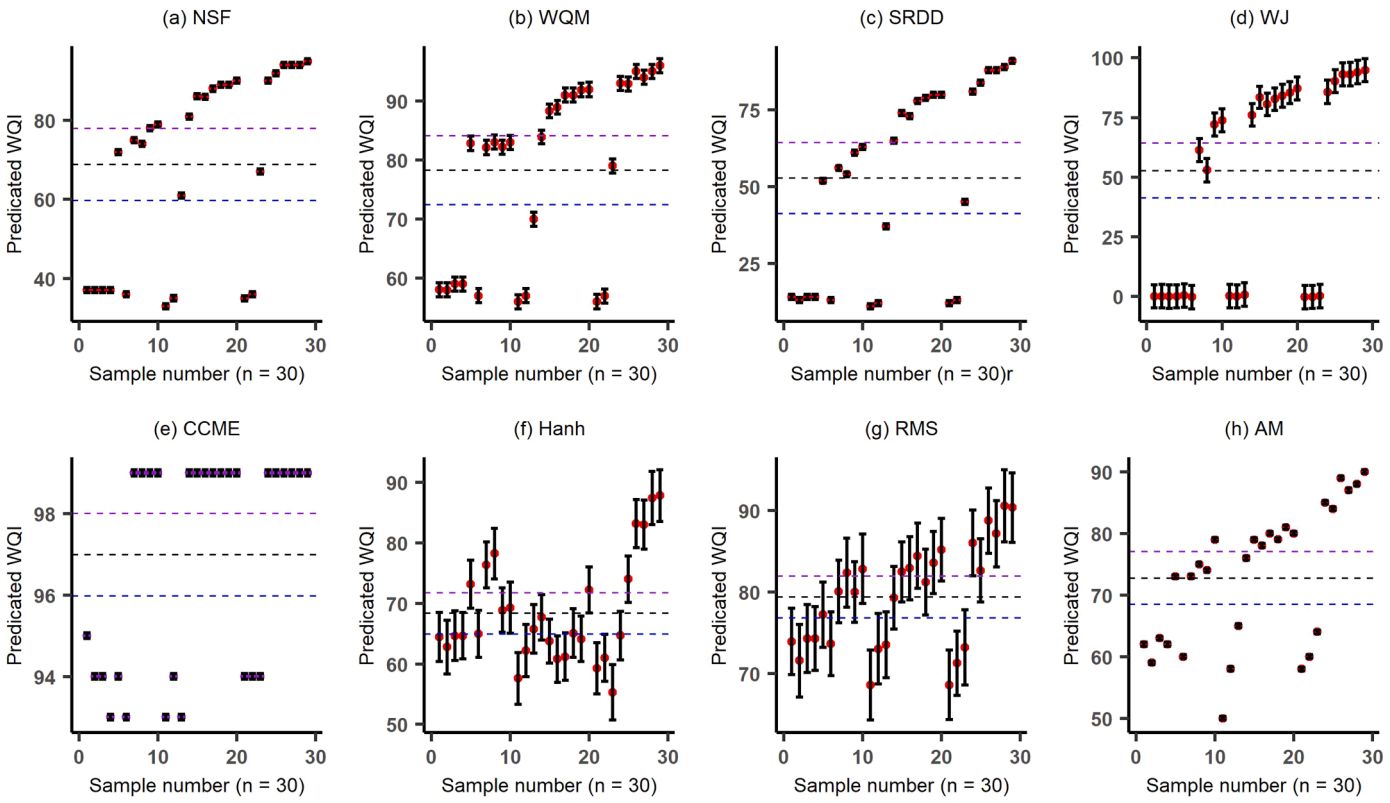


Fig. 14. GPR-predicted WQI values for summer in Cork Harbour. 95% CI bars are shown with various individual GPR means for monitoring site, whereas n=29. A population mean is marked by the black dashed line.

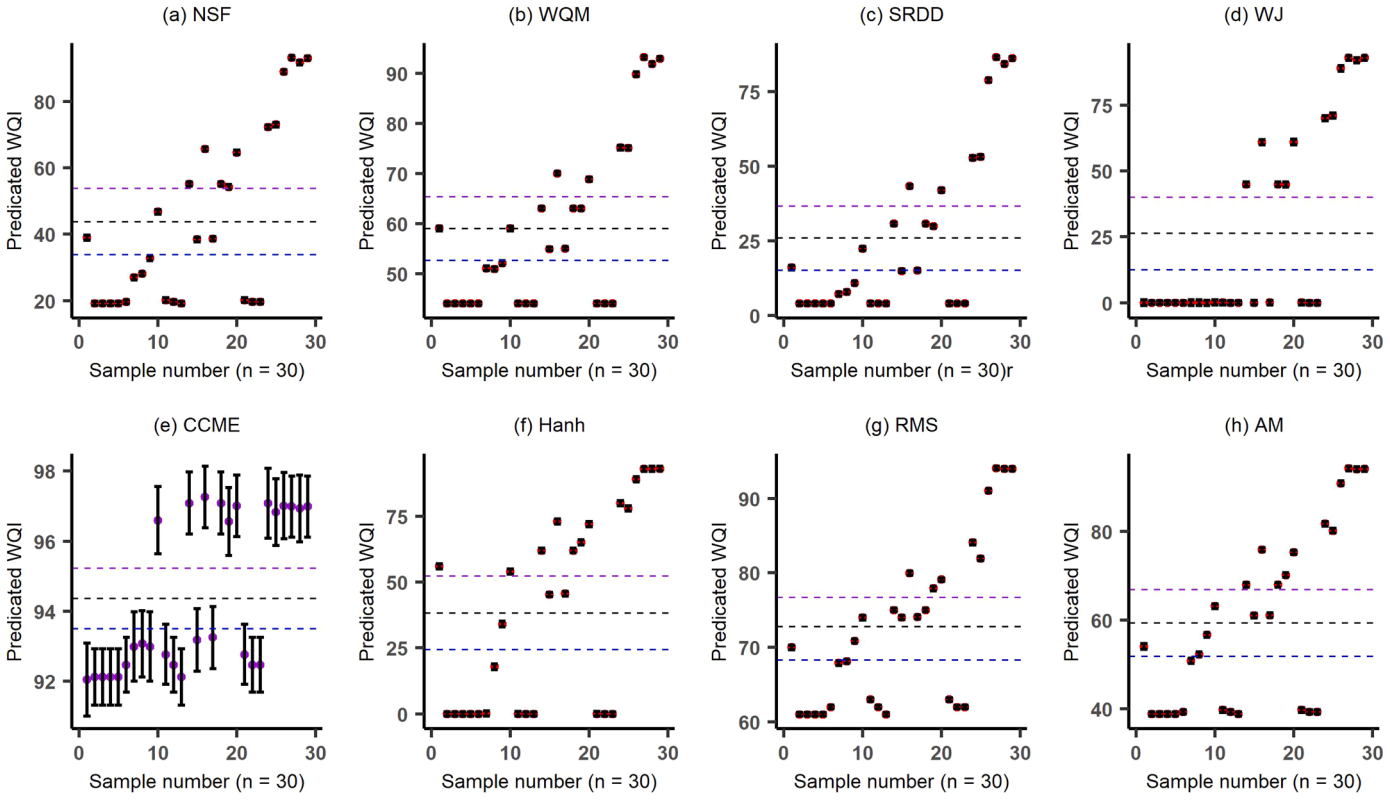


Fig. 15. GPR-predicted WQI values for winter in Cork Harbour. 95% CI bars are shown with various individual GPR means for monitoring site, whereas n=29. A population mean is marked by the black dashed line.

Table 5
t-statistics of GPR predicted WQI values for df = 9999, at 95% confidence level, where p < 0.0001.

WQI models	Summer				Winter			
	t-value	Mean	SE	SD	t	Mean	SE	SD
NSF	417	63.62	0.152	15.27	292	33	0.11	11.34
WQM	**669**	75.0	0.11	11.2	**767**	53.7	0.07	6.99
SRDD	242	43.11	0.18	17.78	134	12.56	0.1	9.4
WJ	47	15.0	0.31	30.95	21	1.73	0.083	8.3
CCME	**9.24E+07**	96.86	0	1.04	**1.25E+06**	94.35	0.0	0.08
Hanh	369.6	62.21	0.17	16.83	52.66	9.05	0.18	17.19
RMS	**1242**	75.0	0.06	6.03	**1082**	71.021	0.07	6.6
AM	**761**	66.32	0.08	8.71	**4304**	58.98	0.01	1.37

*Bold values indicates the higher data variation.

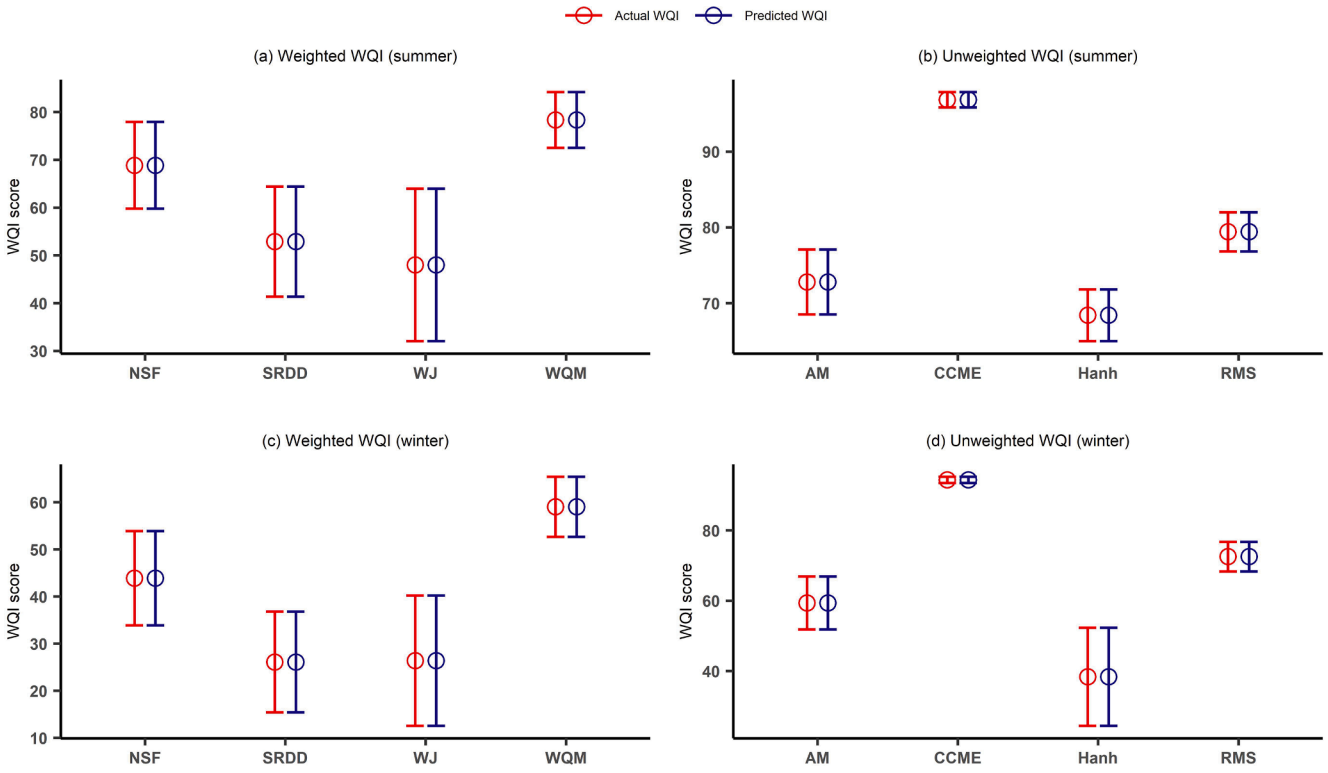


Fig. 16. WQI score means with 95% CI bars for weighted and unweighted WQI models when n is 29, p < 0.0001. "Overlap" refers to the fraction of the average CI error bar arm, i.e., the average of the various WQI model arms. [see Cumming et al. (2007) for the details of CI overlap rule].

were found for the WQM function than for the other three weighted functions (Table 5). For unweighted functions, similar trends were found for the CCME and the RMS models. They also exhibited lower SE and SD values. The results of the predicted uncertainty analysis showed that the WQM and the RMS prediction models could be effective for estimating WQI values in terms of model uncertainty. The findings of prediction uncertainty agree with Uddin et al. (2022a) which found that the SRDD, WJ and CCME models were highly influenced by eclipsing and ambiguity problems.

Throughout the study period (summer and winter), there was no significant difference between the predicted and actual WQI mean, SE, or uncertainty intervals. As can be seen in Fig. 16, there was a strong statistically significant difference among the WQI models at p < 0.0001 in this research. During both seasons, the WQM model had the most consistent WQI variation with the lowest uncertainty and SE trend (Table 5). Similarly, the NSF model had lower SE and shorter intervals than the SRDD and WJ models, which exhibited stronger effects of uncertainty. In terms of seasonal variability of WQI model uncertainty, a significant statistical difference (p < 0.0001) was found for the WQI models between summer and winter seasons (Fig. 16).

Overall, the model intercomparison for GPR-predicted WQI mean, SE, and CI implies that the WQM is the most effective model for assessing coastal water quality in terms of reducing model uncertainty. The ranking of WQI models in terms of generated uncertainty levels, from highest to lowest, is as follows based on the MCS and GPR analyses:

3.5. *WQI model intercomparison*

The Tukey's HSD approach was used for pair-wise analysis to intercompare WQI models. Fig. 17 presents differences between the model means for all combinations of model pairs. There was very little difference between the weighted WQM model and unweighted RMS model for summer period; this is also confirmed by the box plot statistics in Fig. 18a. A small difference was also found for the NSF-Hanh model pair. The largest differences in means were found during the winter period, and this is also confirmed by the boxplot statistics. There was a significant difference (F value for summer = 7127 and winter F = 38675 at p < 0.0001 respectively) in pairwise comparison (p < 0.0001) between weighted and unweighted WQI models.

In terms of individual models, the WQM is significantly better than

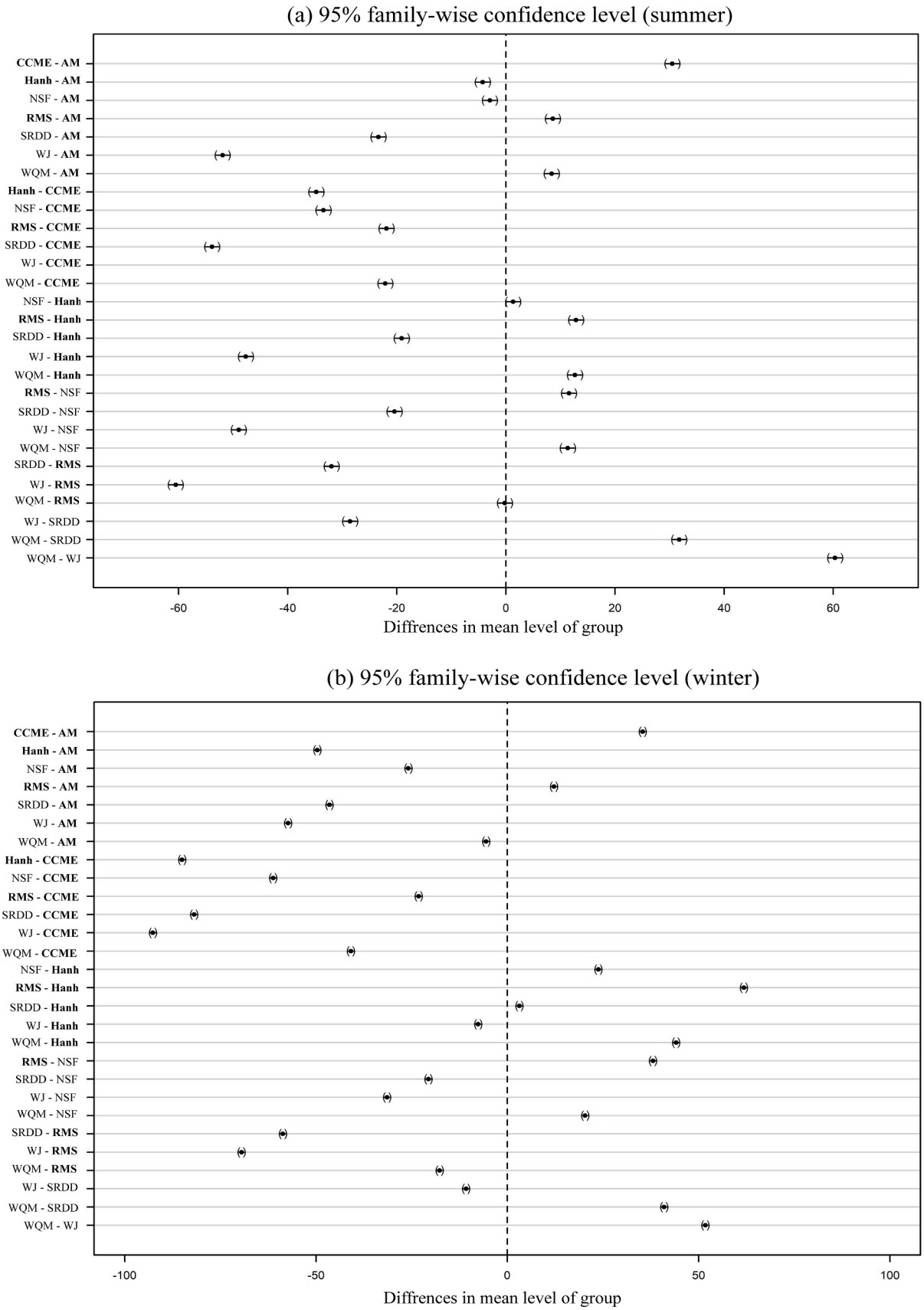


Fig. 17. Pair-wise comparison between the WQI models with 95% CI from Tukey's HSD. The vertical dashed line indicates the point where the difference between the means is equal to zero or the similarity of model statistics where Unweighted WQI models are in bold.

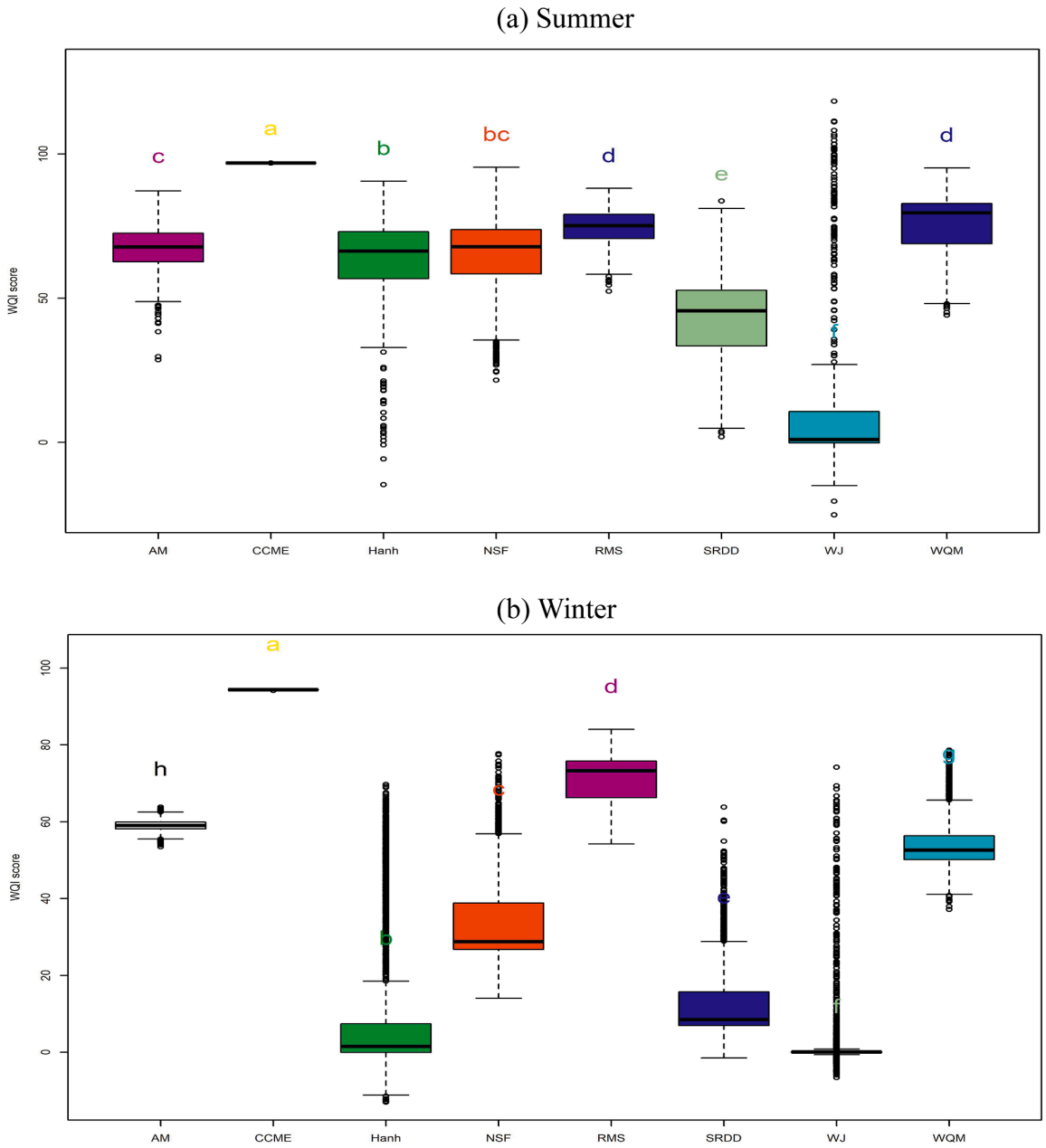


Fig. 18. Box plots of WQI models by group with Tukey's HSD compact letter.

other WQI models during both the summer and the winter period; it was clearly presented in Fig. 18 data obtained from the HSD Tukey test. However, the CCME shows an identical statistical pattern to the other models over the study period (Ahmed et al., 2021; Davies, 2006; Pang et al., 2021). As this study shows that the CCME aggregation function is a source of large uncertainty in the WQI model, and given that it has been shown to be highly susceptible to model eclipsing and ambiguity problems (Uddin et. al., 2022a), the CCME model is not recommended for water quality assessment (Gupta et al., 2003; Lumb et al., 2011; Mladenović-Ranisavljević and Žerajić, 2018). Based on the HSD and uncertainty results, the WQM model appears to be a reliable tool that performed better than other WQI models in this study. This agrees with the findings of Uddin et al. (2022a).

3.6. *Validation of uncertainty results Summary of statistics*

As assessment of the ML predicted scores versus the WQI calculated scores is presented in the Taylor diagrams of Fig. 19. The diagram helps to visually and numerically evaluate model performance using three statistical properties namely: the centred root-mean-square deviation (RMSD), the standard deviation (SD) and the correlation (Calim et al., 2018). The statistics of the calculated and simulated WQI values in Cork Harbour for summer and winter periods are presented in Fig. 19. For the summer period the Hanh, WQM and AM models show the smallest discrepancies and display correlation coefficients of over 95%. In winter, all WQI models cluster around very low SD (<5), low RMSD (<10) and correlation coefficients over 0.8. The CCME model, with correlation coefficients close to 1.0 and near zero RMSD and SD, again shows low discrepancies in WQI scores, but the eclipsing and ambiguity problems have to be emphasized again. With respect to aggregation function, it can be seen from the Taylor statistics that the unweighted WQI models outperform the weighted WQI models.

Overall, the Taylors diagram statistics are consistent with the findings of the MCS and GPR analyses, and reinforce the conclusion that the

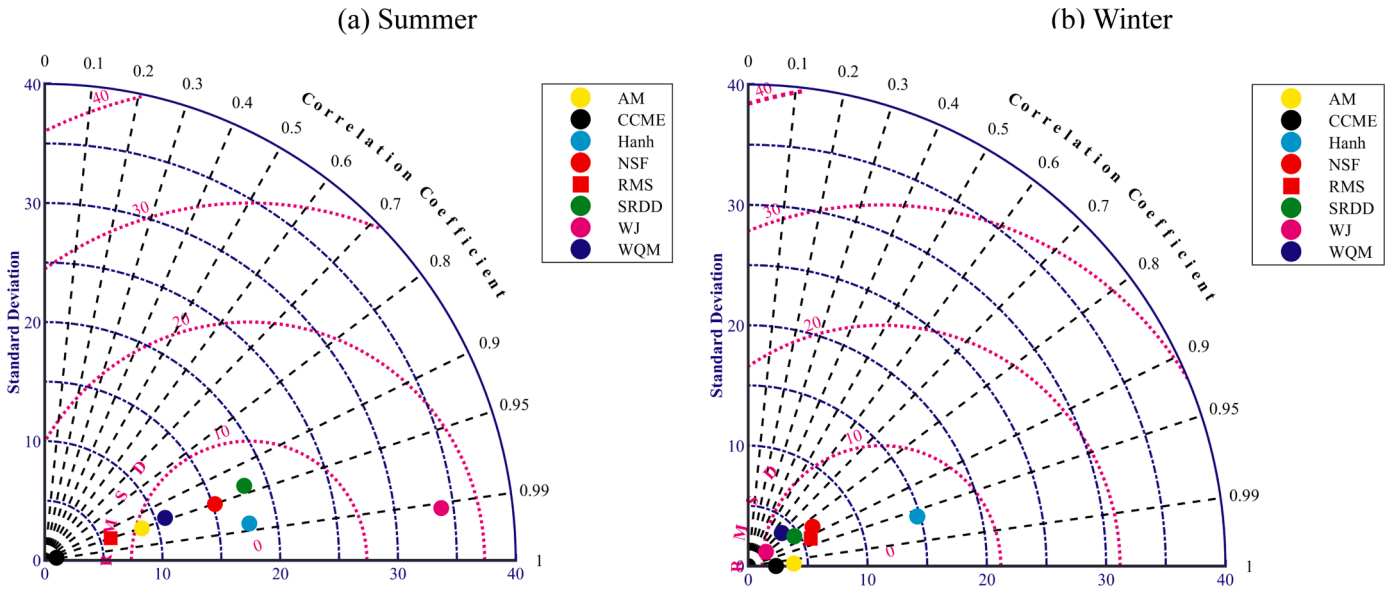


Fig. 19. Taylor diagram statistics for different WQI models.

weighted WQM and unweighted RMS produce lowest uncertainties over the study period. In addition, compared to the weighted methods, the WQM model could be more effective and robust in assessing coastal water quality in terms of the model uncertainty problem.

4. Conclusion

This study is the first attempt to improve the theoretical and experimental understanding of uncertainty in a WQI model. Many researchers argue that uncertainty is produced throughout the various stages of WQI modelling. Most of the recent studies imply that the processes of model indicators selection and sub-indexing play a significant role in generating uncertainty in a WQI model. Some studies suggest that the weighting process also has an effect on the WQI model's efficiency and accuracy. To date, several studies have been conducted which try to evaluate WQI model uncertainty by comparing very generalized statistical measures, but the present study is the first effort to estimate model uncertainty by taking the effect of the individual model stages into account. In this research, a comprehensive quantification of model uncertainty at each step of the WQI model was conducted to understand the individual and combined effects of each contributor of uncertainty in the model. The MCS technique was applied to estimate the WQI model uncertainty, while the GPR algorithm was utilized to predict uncertainties. The major findings of this study can be summarized as follows:

- The values of the selected water quality indicators generated less than 1% of the total model uncertainty; however, the number of input indicators was found to have a significant impact on the model's input uncertainties.
- Sub-index functions were found to contribute significantly to the WQI model uncertainties, and hence affect the model reliability. Therefore, the determination of sub-index functions needs to be taken with care.
- The weighting component produces relatively low levels of uncertainties.
- Significant statistical differences were found between the eight aggregation functions tested. The weighted quadratic mean (WQM) function was found to provide a plausible assessment of the WQ of coastal waters at reduced uncertainty levels. The findings of this study also suggest that the unweighted root mean squared (RMS) aggregation function could also be potentially used for assessment of coastal WQ.
- Finally, the WQM and RMS models were found to exhibit a higher prediction accuracy and these WQI models could be effective and reliable tools to provide low-uncertainty assessment of water quality status.

This research provides a robust statistically sound methodology for assessment of WQI model uncertainties. A set of statistical tools in this method can be used to identify sources of uncertainty, quantify them and evaluate their impact on the overall model performance. Therefore, the finding of this research could aid a range of stakeholders including decision-makers, researchers, and agencies responsible for water quality monitoring, assessment and management. The methodology was only tested for one domain application. For its further improvement and validation it should be applied to various types of waterbodies.

Declaration of Competing Interest

The authors state that they did not experience competing financial interests or personal relationships that could have been influenced by the work reported in this paper.

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Supplementary materials

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References

Abbasi, T., Abbasi, S.A., 2012. *Water quality indices*. Elsevier. https://doi.org/10.1016/C2010-0-69472-7.

Abdar, M., Pourpanah, F., Hussain, S., Rezazadegan, D., Liu, L., Ghavamzadeh, M., Fieguth, P., Cao, X., Khosravi, A., Acharya, U.R., Makarenkov, V., Nahavandi, S., 2021. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. Inf. Fusion 76, 243–297. https://doi.org/10.1016/j.inffus.2021.05.008.

Ahmed, M., Mumtaz, R., Zaidi, S.M.H., 2021. Analysis of water quality indices and machine learning techniques for rating water pollution: A case study of Rawal Dam, Pakistan. Water Supply 21, 3225–3250. https://doi.org/10.2166/ws.2021.082.

Asadollah, S.B.H.S., Sharafati, A., Motta, D., Yaseen, Z.M., 2021. River water quality index prediction and uncertainty analysis: A comparative study of machine learning models. J. Environ. Chem. Eng. 9, 104599 https://doi.org/10.1016/j.jece.2020.104599.

BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, OIML, 2020. GUM modeling: Guide to the expression of uncertainty in measurement - Part 6: Developing and using measurement models JCGM GUM-6, pp. 1–103.

Bui, D.T., Khosravi, K., Tiefenbacher, J., Nguyen, H., Kazakis, N., 2020. Improving prediction of water quality indices using novel hybrid machine-learning algorithms. Sci. Total Environ. 721, 137612 https://doi.org/10.1016/j.scitotenv.2020.137612.

Bode, G., Thul, S., Baranski, M., Müller, D., 2020. Real-world application of machine-learning-based fault detection trained with experimental data. Energy 198. https://doi.org/10.1016/j.energy.2020.117323.

Calim, M.C., Nobre, P., Oke, P., Schiller, A., Siqueira, L.S.P., Castelão, G.P., 2018. A new tool for model assessment in the frequency domain – Spectral Taylor Diagram : application to a global ocean general circulation model with tides. Geosci. Model Dev. https://doi.org/10.5194/gmd-2018-5.

Carsten Von Der Ohe, P., Prüß, A., Schäfer, R.B., Liess, M., de Deckere, E., Brack, W., 2007. Water quality indices across Europe - A comparison of the good ecological status of five river basins. J. Environ. Monitor. 9, 970–978. https://doi.org/10.1039/b704699p.

Cumming, G., Fidler, F., Vaux, D.L., 2007. Error bars in experimental biology. J. Cell Biol. 177, 7–11. https://doi.org/10.1083/jcb.200611141.

Davies, J.M., 2006. Application and tests of the canadian water quality index for assessing changes in water quality in lakes and rivers of central North America. Lake Reserv. Manag. 22, 308–320. https://doi.org/10.1080/07438140609354365.

Ekici, S., Unal, F., Ozleyen, U., 2019. Comparison of different regression models to estimate fault location on hybrid power systems. IET Gener. Transm. Distrib. 13, 4756–4765. https://doi.org/10.1049/iet-gtd.2018.6213.

Elgeldawi, E., Sayed, A., Galal, A.R., Zaki, A.M., 2021. Hyperparameter tuning for machine learning algorithms used for arabic sentiment analysis. Informatics 8, 1–21. https://doi.org/10.3390/informatics8040079.

Farrance, I., Frenkel, R., 2012. Uncertainty of measurement: A review of the rules for calculating Uncertainty components through functional relationships. Clin. Biochem. Rev. 33, 49–75.

Fernandes, M., Corchado, J.M., Marreiros, G., 2022. Machine learning techniques applied to mechanical fault diagnosis and fault prognosis in the context of real industrial manufacturing use-cases: a systematic literature review. Appl. Intell. https://doi.org/10.1007/s10489-022-03344-3.

Galal Uddin, M., Moniruzzaman, M., Khan, M., 2017. Evaluation of Groundwater Quality Using CCME Water Quality Index in the Rooppur Nuclear Power Plant Area, Ishwardi, Pabna, Bangladesh. Am. J. Environ. Prot. https://doi.org/10.12691/env-5-2-2.

Grbić, R., Kurtagić, D., Slišković, D., 2013. Stream water temperature prediction based on Gaussian process regression. Expert Syst. Appl. 40, 7407–7414. https://doi.org/10.1016/j.eswa.2013.06.077.

Guimaraes Couto, P.R., Carreteiro, J., de Oliveir, S.P.. Monte Carlo Simulations Applied to Uncertainty in Measurement, in: Theory and Applications of Monte Carlo Simulations. InTech. https://doi.org/10.5772/53014.

Gupta, A.K., Gupta, S.K., Patil, R.S., 2003. A comparison of water quality indices for coastal water. J. Environ. Sci. Health A Tox. Hazard. Subst. Environ. Eng. 38, 2711–2725. https://doi.org/10.1081/ESE-120024458.

Goeman, J.J., Solari, A., 2021. Comparing Three Groups _ Enhanced Reader.pdf. Am. Stat. https://doi.org/10.1080/00031305.2021.2002188

Haghiabi, A.H., Nasrolahi, A.H., Parsaie, A., 2018. Water quality prediction using machine learning methods. Water Qual. Res. J. 53, 3–13. https://doi.org/10.2166/wqrj.2018.025.

Hartnett, M., Nash, S., 2015. An integrated measurement and modeling methodology for estuarine water quality management. Water Sci. Eng. 8, 9–19. https://doi.org/10.1016/j.wse.2014.10.001.

Jafari, F., Javadi, S., Golmohammadi, G., Mohammadi, K., Khodadadi, A., Mohammadzadeh, M., 2016. Groundwater risk mapping prediction using mathematical modeling and the Monte Carlo technique. Environ. Earth Sci. 75, 1–11. https://doi.org/10.1007/s12665-016-5335-9.

Jiang, Y., Li, C., Sun, L., Guo, D., Zhang, Y., Wang, W., 2021. A deep learning algorithm for multi-source data fusion to predict water quality of urban sewer networks. J. Clean. Prod. 318 https://doi.org/10.1016/j.jclepro.2021.128533.

Jiang, L., Li, Y., Zhao, X., Tillotson, M.R., Wang, W., Zhang, S., Sarpong, L., Asmaa, Q., Pan, B., 2018. Parameter uncertainty and sensitivity analysis of water quality model in Lake Taihu, China. Ecol. Modell. 375, 1–12. https://doi.org/10.1016/j.ecolmodel.2018.02.014.

Joint Committee For Guides In Metrology, 2008. Evaluation of measurement data — Guide to the expression of uncertainty in measurement. Int. Organ. Stand. Geneva ISBN 50, 134. https://doi.org/10.1373/clinchem.2003.030528.

Juwana, I., Muttil, N., Perera, B.J.C., 2016. Uncertainty and sensitivity analysis of West Java Water Sustainability Index - A case study on Citarum catchment in Indonesia. Ecol. Indic. 61, 170–178. https://doi.org/10.1016/j.ecolind.2015.08.034.

Khoi, D.N., Quan, N.T., Linh, D.Q., Nhi, P.T.T., Thuy, N.T.D., 2022. Using Machine Learning Models for Predicting the Water Quality Index in the La Buong River, Vietnam. Water (Switzerland) 14. https://doi.org/10.3390/w14101552.

Liang, S., Jia, H., Xu, C., Xu, T., Melching, C., 2016. A Bayesian approach for evaluation of the effect of water quality model parameter uncertainty on TMDLs: A case study of Miyun Reservoir. Sci. Total Environ. 560–561, 44–54. https://doi.org/10.1016/j.scitotenv.2016.04.001.

Liu, Y., Yang, P., Hu, C., Guo, H., 2008. Water quality modeling for load reduction under uncertainty: A Bayesian approach. Water Res. 42, 3305–3314. https://doi.org/10.1016/j.watres.2008.04.007.

Loucks, P.D., van Beek, E., 1981. Water resource systems planning and analysis - An introduction to Methods, Models, and Applications. Adv. Water Resour.

Lumb, A., Sharma, T.C., Bibeault, J.-F., Klawunn, P., 2011. A Comparative Study of USA and Canadian Water Quality Index Models. Water Qual. Expo. Health 3, 203–216. https://doi.org/10.1007/s12403-011-0056-5.

Meyer, V.R., 2007. Measurement uncertainty. J. Chromatogr. A 1158, 15–24. https://doi.org/10.1016/j.chroma.2007.02.082.

Mishra, S., 2009. Uncertainty and sensitivity analysis techniques for hydrologic modeling. J. Hydroinformat. 11, 282–296. https://doi.org/10.2166/hydro.2009.048.

Mladenović-Ranisavljević, I.I., Žerajić, S.A., 2018. Comparison of different models of water quality index in the assessment of surface water quality. Int. J. Environ. Sci. Tech. 15, 665–674. https://doi.org/10.1007/s13762-017-1426-8.

Nash, S., Hartnett, M., Dabrowski, T., 2011. Modelling phytoplankton dynamics in a complex estuarine system. Proc. Inst. Civ. Eng., Water marit. eng. 164, 35–54. https://doi.org/10.1680/wama.800087.

Ng, A.W.M., Perera, B.J.C., 2001. Uncertainty and sensitivity analysis of river water quality model parameters. Prog. Water Resour. 48, 175–184.

Pang, M., Xu, R., Hu, Z., Wang, J., Wang, Y., 2021. Uncertainty and sensitivity analysis of input conditions in a large shallow lake based on the latin hypercube sampling and morris methods. Water (Switzerland) 13. https://doi.org/10.3390/w13131861.

Parween, S., Siddique, N.A., Mahammad Diganta, M.T., Olbert, A.I., Uddin, Md.G., 2022. Assessment of urban river water quality using modified NSF water quality index model at Siliguri city, West Bengal, India. Environ. Sustain. Indic. 16, 100202. https://doi.org/10.1016/j.indic.2022.100202.

Rahman, A., 2019. Statistics-based data preprocessing methods and machine learning algorithms for big data analysis. Int. J. Artif. Intell. 17 (2), 44–65.

Rahman, A., Harding, A., Tanton, R., Liu, S., 2013. Simulating the Characteristics of Populations at the Small Area Level: New Validation Techniques for a Spatial Microsimulation Model in Australia. Comput. Stat. Data. Anal. 57 (1), 149–165. https://doi.org/10.1016/j.csda.2012.06.018.

Rahman, A., Harding, A., 2017. Small area estimation and microsimulation modeling. CRC Press. https://doi.org/10.1201/9781315372143.

Rodríguez, E., Arqués, J.L., Rodríguez, R., Nuñez, M., Medina, M., Talarico, T.L., Casas, I. A., Chung, T.C., Dobrogosz, W.J., Axelsson, L., Lindgren, S.E., Dobrogosz, W.J., Kerkeni, L., Ruano, P., Delgado, L.L., Picco, S., Villegas, L., Tonelli, F., Merlo, M., Rigau, J., Diaz, D., Masuelli, M., 1989. Monte Carlo Simulations Applied to Uncertainty in Measurement. Intech 32, 137–144.

Rouder, J.N., Engelhardt, C.R., McCabe, S., Morey, R.D., 2016. Model comparison in ANOVA. Psychon. Bull. Rev. 23, 1779–1786. https://doi.org/10.3758/s13423-016-1026-5.

Sankaran, M., Sarkar, S., 2009. Uncertainty analysis methods. Issues Quant. Qual. Risk Model. Appl. Import Risk Assess. ACERA Proj. 1–26.

Saufi, S.R., Ahmad, Z.A.bin, Leong, M.S., Lim, M.H., 2019. Challenges and opportunities of deep learning models for machinery fault detection and diagnosis: A review. IEEE Access. https://doi.org/10.1109/ACCESS.2019.2938227.

Seifi, A., Dehghani, M., Singh, V.P., 2020. Uncertainty analysis of water quality index (WQI) for groundwater quality evaluation: Application of Monte-Carlo method for weight allocation. Ecol. Indic. 117, 106653 https://doi.org/10.1016/j.ecolind.2020.106653.

Shekar, B.H., Dagnew, G., 2019. Grid search-based hyperparameter tuning and classification of microarray cancer data. In: 2019 2nd International Conference on Advanced Computational and Communication Paradigms, ICACCP 2019. https://doi.org/10.1109/ICACCP.2019.8882943.

Shaw, B.D., 2017. Uncertainty Analysis of Experimental Data with R. Chapman and Hall/CRC. CRC Press, Boca Raton. https://doi.org/10.1201/9781315366715, 2017.

Sutadian, A.D., Muttil, N., Yilmaz, A.G., Perera, B.J.C., 2018. Development of a water quality index for rivers in West Java Province, Indonesia. Ecol. Indic. https://doi.org/10.1016/j.ecolind.2017.11.049.

Sutadian, A.D., Muttil, N., Yilmaz, A.G., Perera, B.J.C., 2017. Using the Analytic Hierarchy Process to identify parameter weights for developing a water quality index. Ecol. Indic. 75, 220–233. https://doi.org/10.1016/j.ecolind.2016.12.043.

Smith, D.G., 1990. A better water quality indexing system for rivers and streams. Water Res. 24, 1237–1244. https://doi.org/10.1016/0043-1354(90)90047-A.

Sutadian, A.D., Muttil, N., Yilmaz, A.G., Perera, B.J.C., 2016. Development of river water quality indices—a review. Environ. Monit. Assess. https://doi.org/10.1007/s10661-015-5050-0.

Tavazza, F., Decost, B., Choudhary, K., 2021. Uncertainty prediction for machine learning models of material properties. ACS Omega 6, 32431–32440. https://doi.org/10.1021/acsomega.1c03752.

Uddin, M.G., Nash, S., Olbert, A.I., 2021. A review of water quality index models and their use for assessing surface water quality. Ecol. Indic. https://doi.org/10.1016/j.ecolind.2020.107218.

Uddin, M.G., Nash, S., Rahman, A., Olbert, A.I., 2022a. A comprehensive method for improvement of water quality index (WQI) models for coastal water quality assessment. Water Res 219, 118532. https://doi.org/10.1016/j.watres.2022.118532.

Uddin, M.G., Nash, S., Rahman, A., Olbert, A.I., 2022b. Development of a water quality index model - a comparative analysis of various weighting methods. In: Çiner, P.D.A. (Ed.), Mediterranean Geosciences Union Annual Meeting (MedGU-21). Istanbul, pp. 1–6.

Uddin, M.G., Nash, S., Rahman, A., Diganta, M.T.M., Olbert, A.I., 2022c. Robust Machine Learning Algorithms for Predicting Coastal Water Quality Index. J Environ Manage 321, 115923. https://doi.org/10.1016/j.jenvman.2022.115923.

Uddin, G., Nash, S., Rahman, A., Olbert, A.I., 2023. Performance analysis of the water quality index model for predicting water state using machine learning techniques. Process Saf. Environ. Prot. 169 https://doi.org/10.1016/j.psep.2022.11.073.

Uddin, M.G., Olbert, A.I., Nash, S., 2020a. Assessment of water quality using Water Quality Index (WQI) models and advanced geostatistical technique. In: Civil Engineering Research Association of Ireland (CERAI). Civil Engineering Research Association of Ireland (CERAI). Cork Institute of Technology, Cork, pp. 594–599.

Uddin, M.G., Stephen, Nash, Olbert, A.I., 2020b. Application of Water Quality Index Models to an Irish Estuary, in: Civil and Environmental Research. Civil Engineering Research Association of Ireland (CERAI). Civil Engineering Research Association of Ireland (CERAI). Cork Institute of Technology, Cork, pp. 576–581.

Wang, L., Del Fiol, G., Bray, B.E., Haug, P.J., 2017. Generating disease-pertinent treatment vocabularies from MEDLINE citations. J. Biomed. Inform. 65, 46–57. https://doi.org/10.1016/j.jbi.2016.11.004.

Wu, Z., Lai, X., Li, K., 2021. Water quality assessment of rivers in Lake Chaohu Basin (China) using water quality index. Ecol. Indic. 121 https://doi.org/10.1016/j.ecolind.2020.107021.

Wu, J., Zou, R., Yu, S.L., 2006. Uncertainty analysis for coupled watershed and water quality modeling systems. J. Water Resour. Plan. Manag. https://doi.org/10.1061/(ASCE)0733-9496(2006)132:5(351).

Xiong, Z., Cui, Y., Liu, Z., Zhao, Y., Hu, M., Hu, J., 2020. Evaluating explorative prediction power of machine learning algorithms for materials discovery using k-fold forward cross-validation. Comput. Mater. Sci. 171, 109203 https://doi.org/10.1016/j.commatsci.2019.109203.

Xu, X., Lai, T., Jahan, S., Farid, F., Bello, A., 2022. A Machine Learning Predictive Model to Detect Water Quality and Pollution. Future Internet 14, 324. https://doi.org/10.3390/fi14110324.

Yang, C., Liu, J., Zeng, Y., Xie, G., 2019. Real-time condition monitoring and fault detection of components based on machine-learning reconstruction model. Renew. Energy 133, 433–441. https://doi.org/10.1016/j.renene.2018.10.062.

Zhang, Z., Ye, L., Qin, H., Liu, Y., Wang, C., Yu, X., Yin, X., Li, J., 2019. Wind speed prediction method using Shared Weight Long Short-Term Memory Network and Gaussian Process Regression. Appl. Energy 247, 270–284. https://doi.org/10.1016/j.apenergy.2019.04.047.

Zhao, X., Shen, Z.Y., Xiong, M., Qi, J., 2011. Key uncertainty sources analysis of water quality model using the first order error method. Int. J. Environ. Sci. Technol. 8, 137–148. https://doi.org/10.1007/BF03326203.

Zhu, M., Wang, J., Yang, X., Zhang, Y., Zhang, L., Ren, H., Wu, B., Ye, L., 2022. A review of the application of machine learning in water quality evaluation. Eco-Environ. Health 1, 107–116. https://doi.org/10.1016/j.eehl.2022.06.001.

Zotou, I., Tsihrintzis, V.A., Gikas, G.D., 2019. Performance of Seven Water Quality Indices (WQIs) in a Mediterranean River. Environ. Monit. Assess. 191 https://doi.org/10.1007/s10661-019-7652-4.

Rasmussen, C.E., 2004. Gaussian Processes in Machine Learning, in: Bousquet, O., von Luxburg, U., Rätsch, G. (Eds.), Advanced Lectures on Machine Learning: ML Summer Schools 2003, Canberra, Australia, February 2 - 14, 2003, Tübingen, Germany, August 4 - 16, 2003, Revised Lectures. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 63–71. https://doi.org/10.1007/978-3-540-28650-9_4.